THE UNIVERSITY of EDINBURGH

Connecting scales in the SMEFT at the LHC and future colliders

DAMTP seminar

Based on arXiv: 2502.20453, 2504.05974

In collaboration with Luca Mantani, Juan Rojo,
Alejo Rossia, Eleni Vryonidou

Jaco ter Hoeve

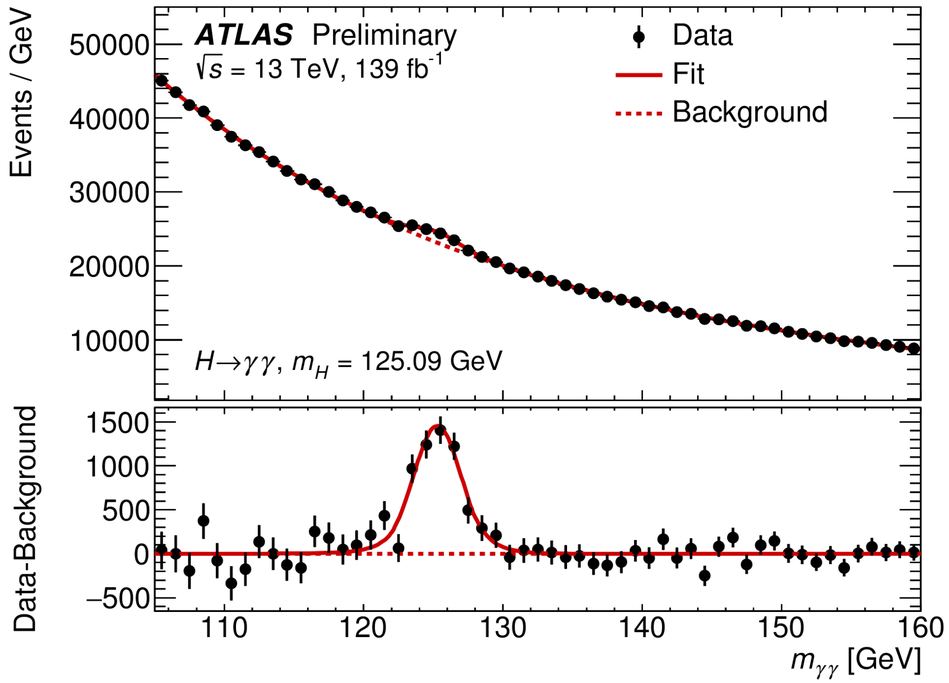
02/05/25

This talk

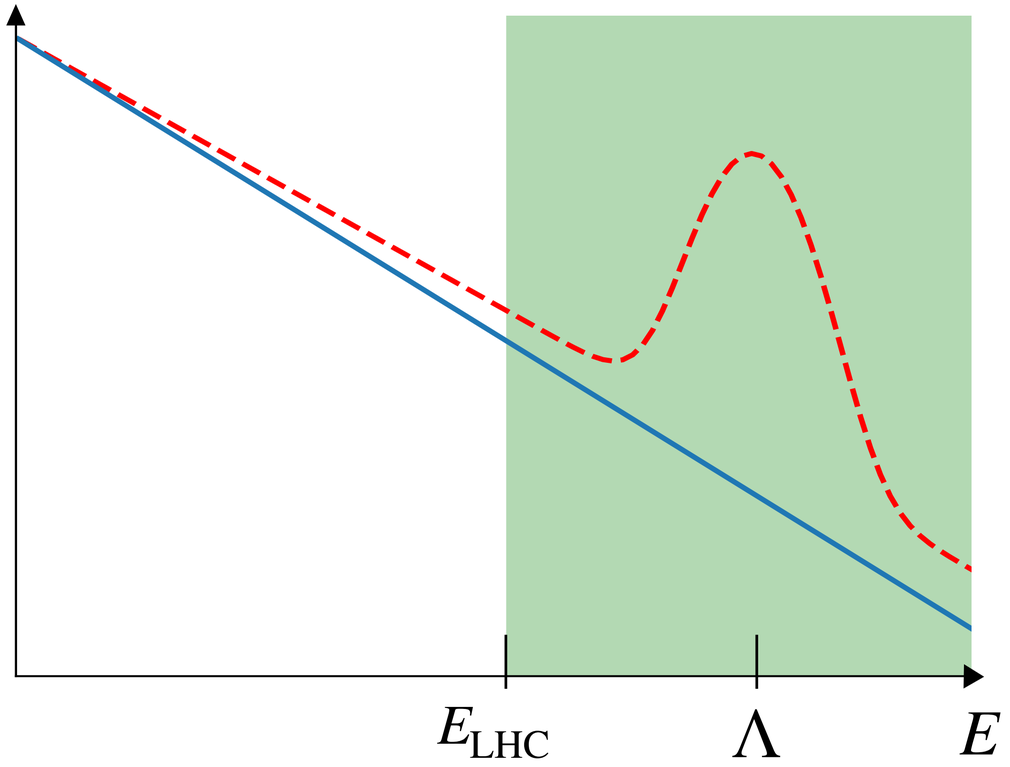
- A brief introduction to the SMEFT
- Renormalisation Group Evolution in global SMEFT fits
- The SMEFT at HL-LHC and FCC-ee
- Constraining the Higgs self-coupling
- Summary

How to look for New Physics

1. Directly: bump hunting

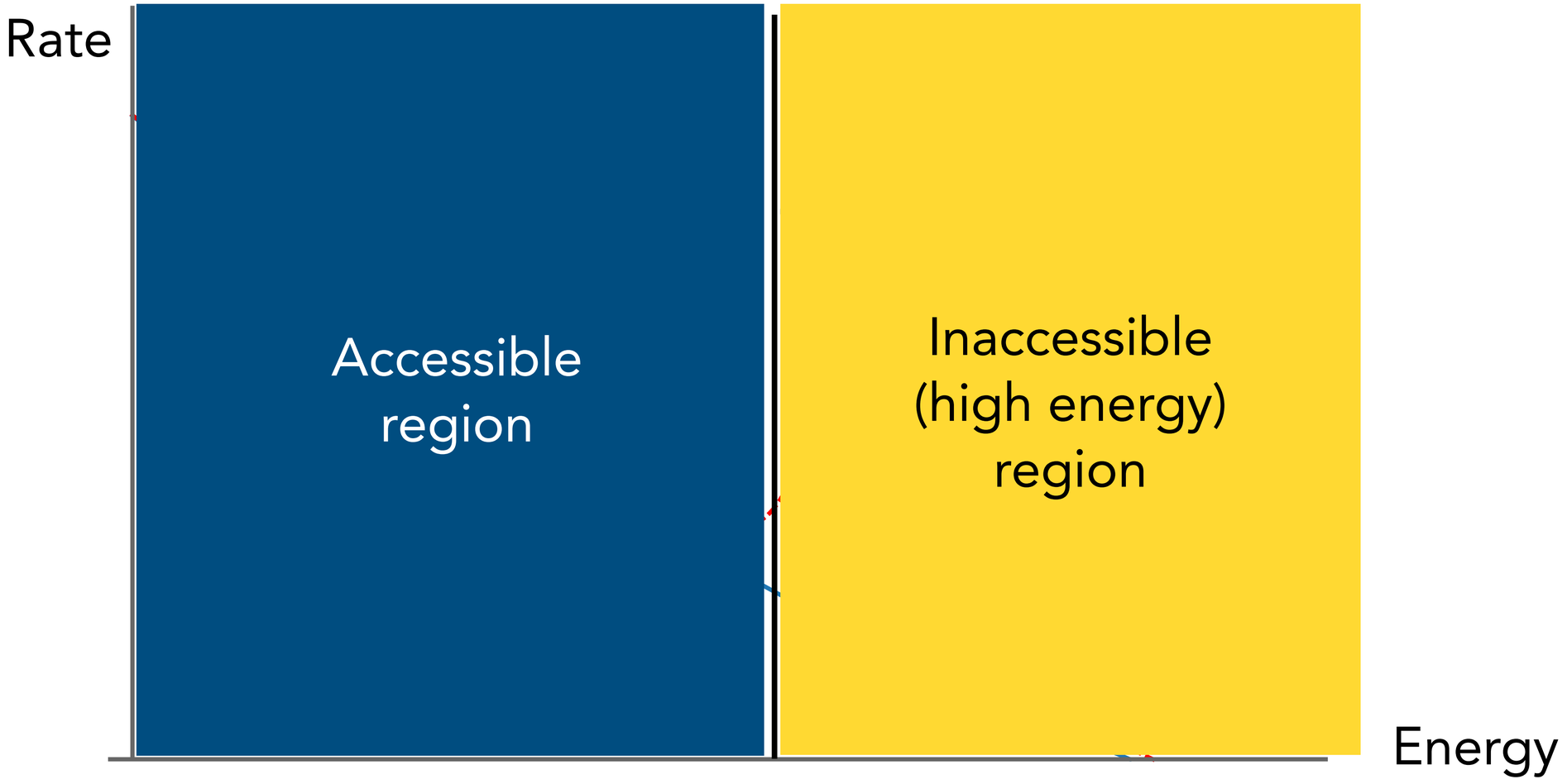


2. Indirectly: tiny deviations in tails



The Standard Model Effective Field Theory

Brivio, Trott, arXiv: 1706.08945

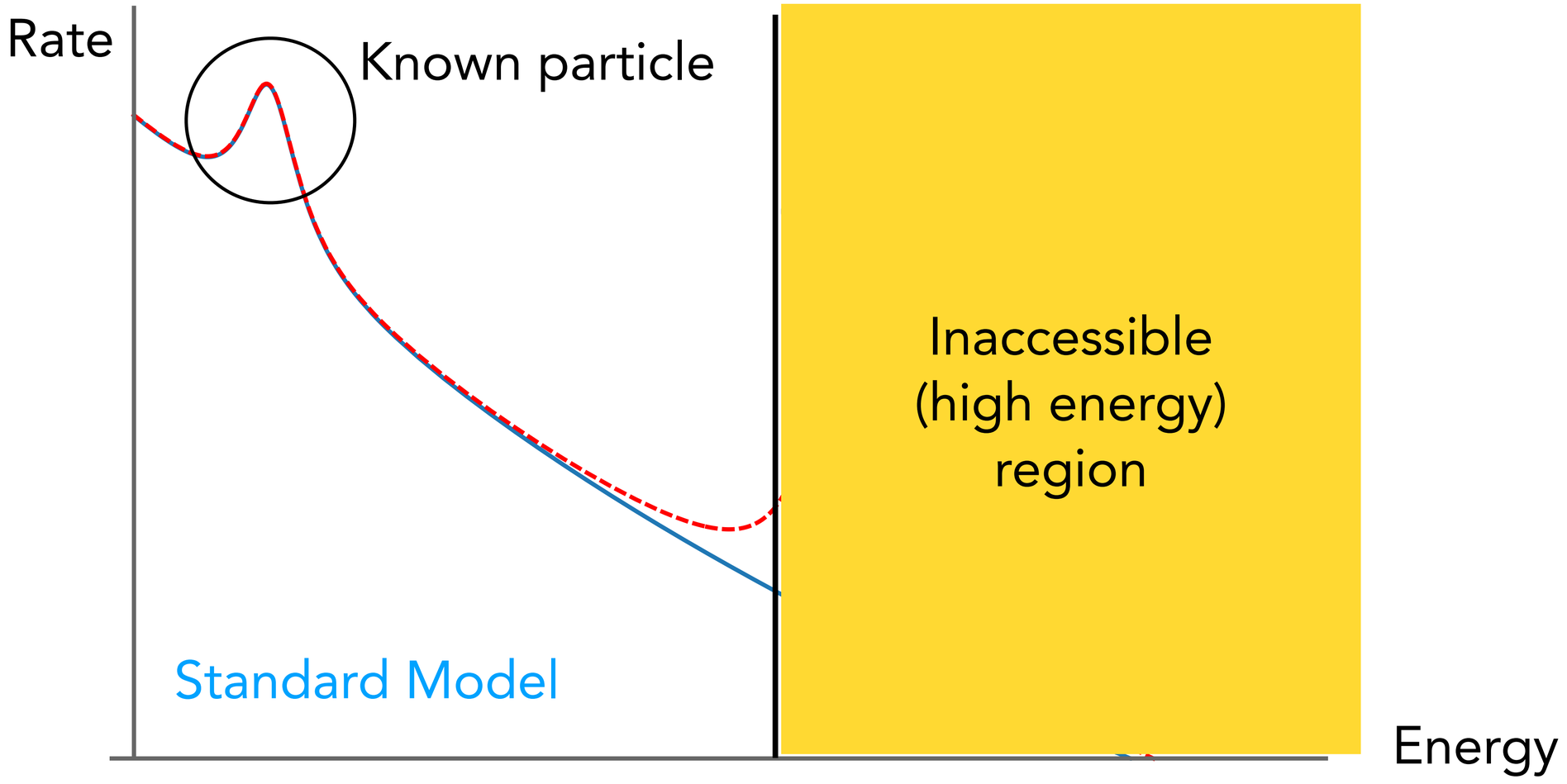


Adapted from E. Vryonidou

The SMEFT: the way to probe new physics beyond the direct collider energy reach

The Standard Model Effective Field Theory

Brivio, Trott, arXiv: 1706.08945

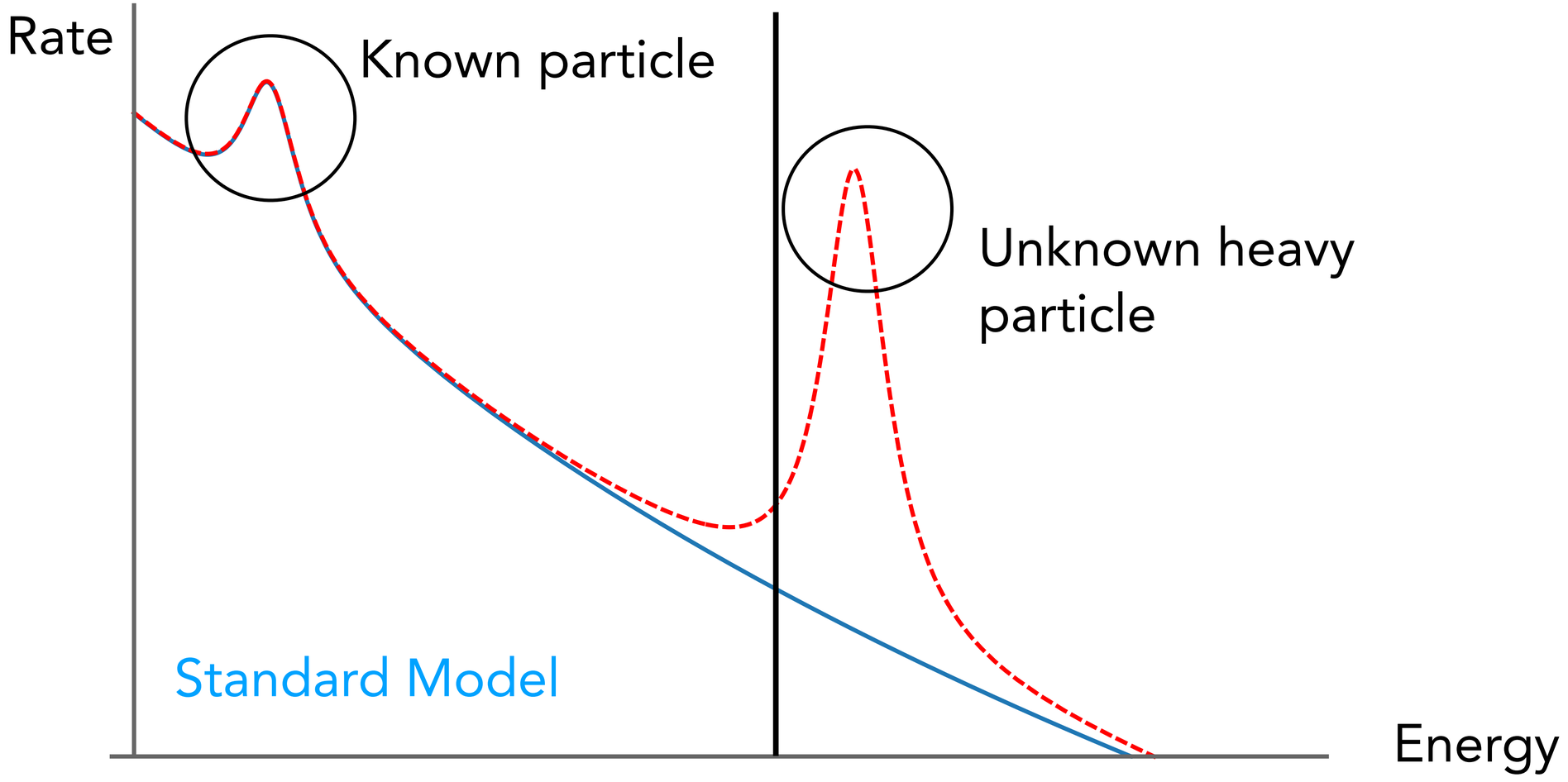


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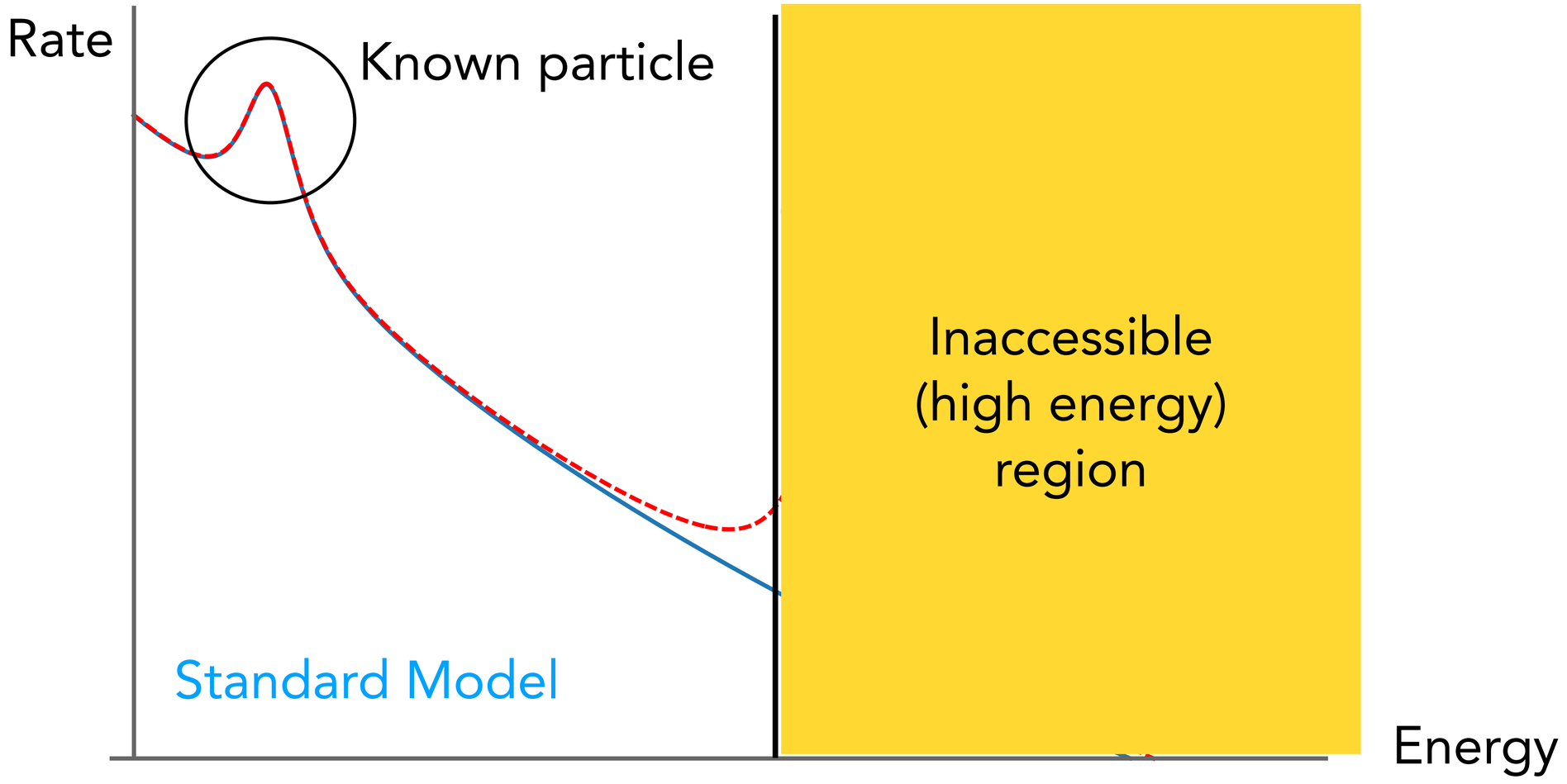


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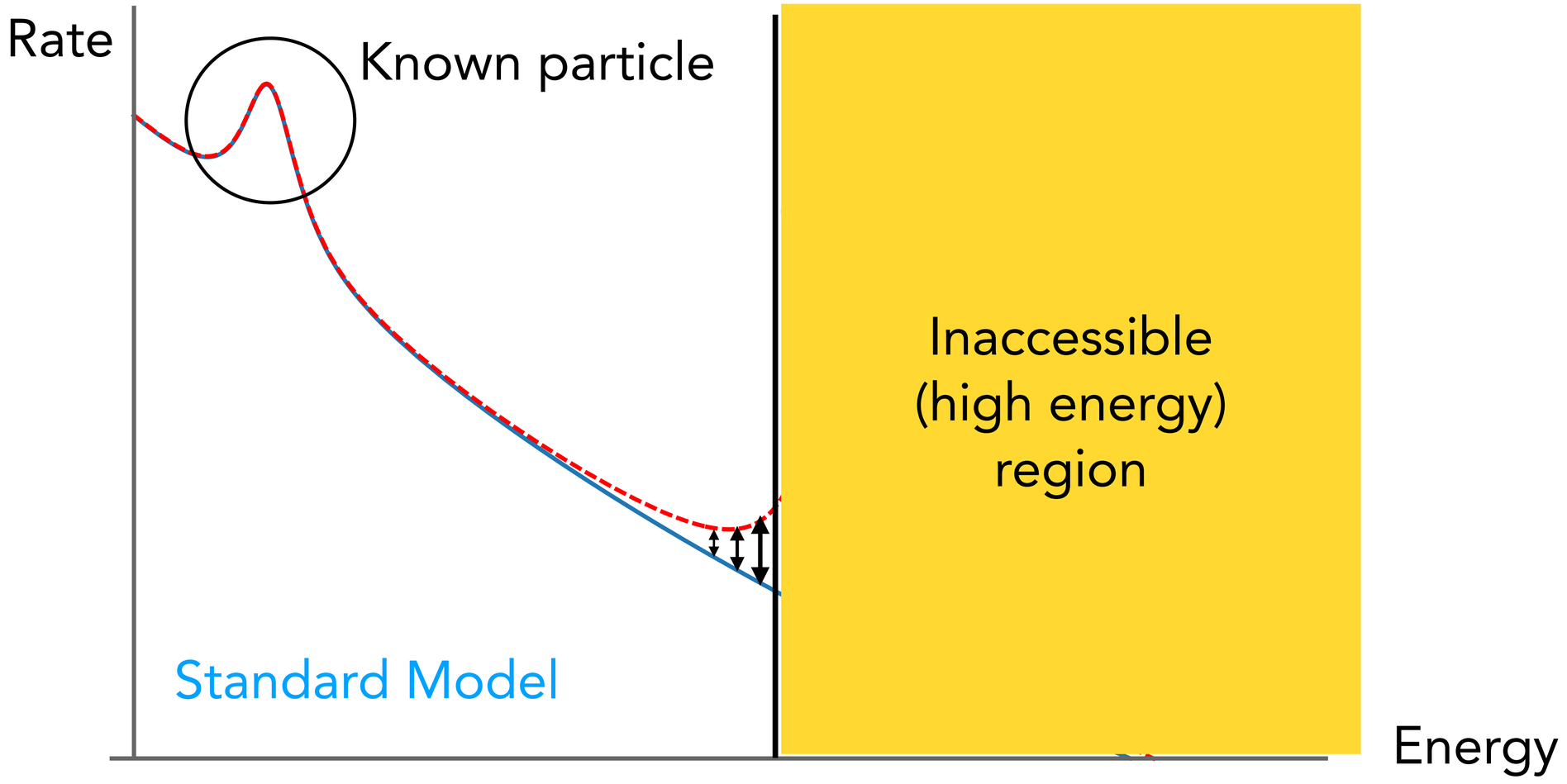


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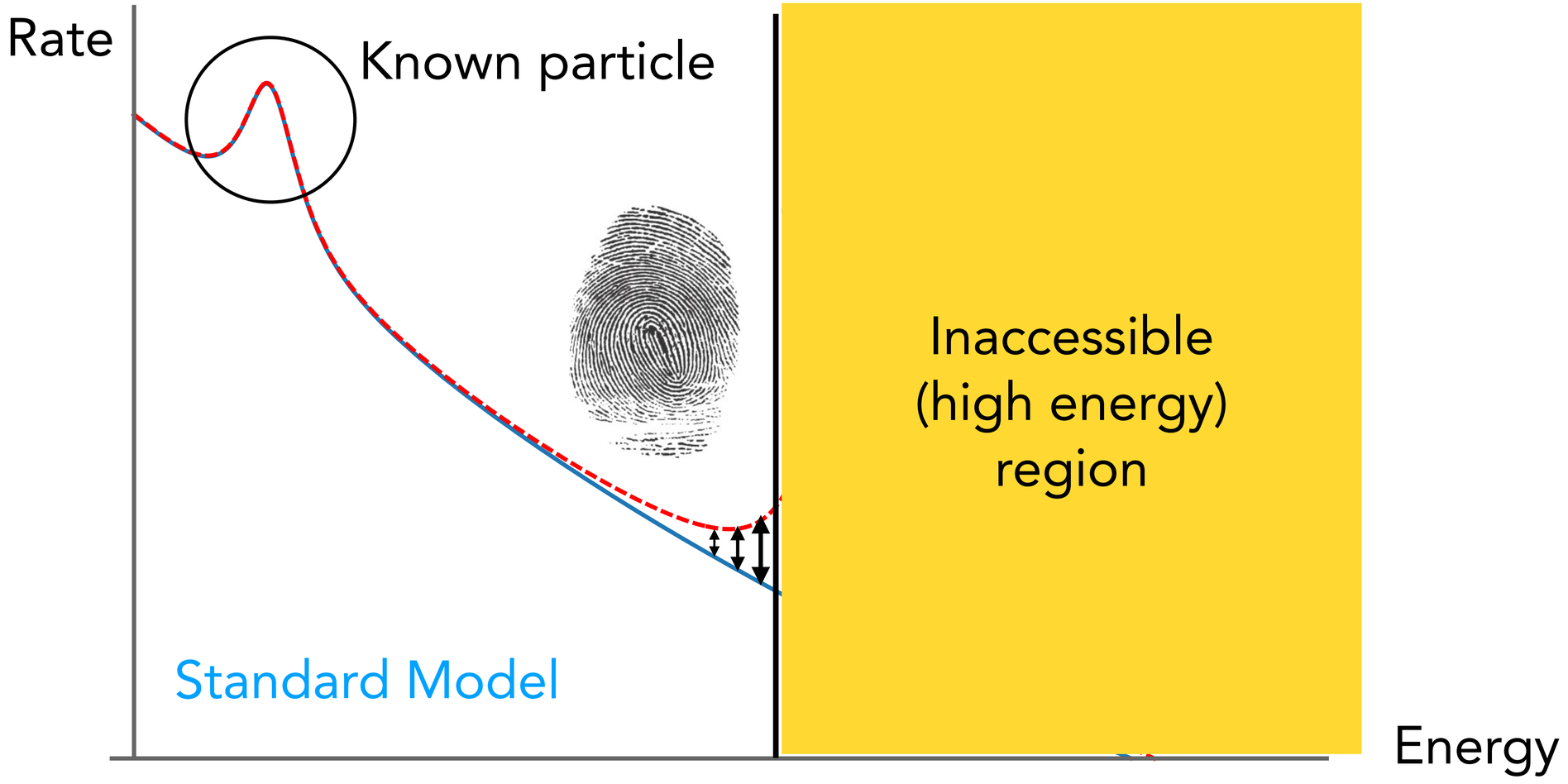


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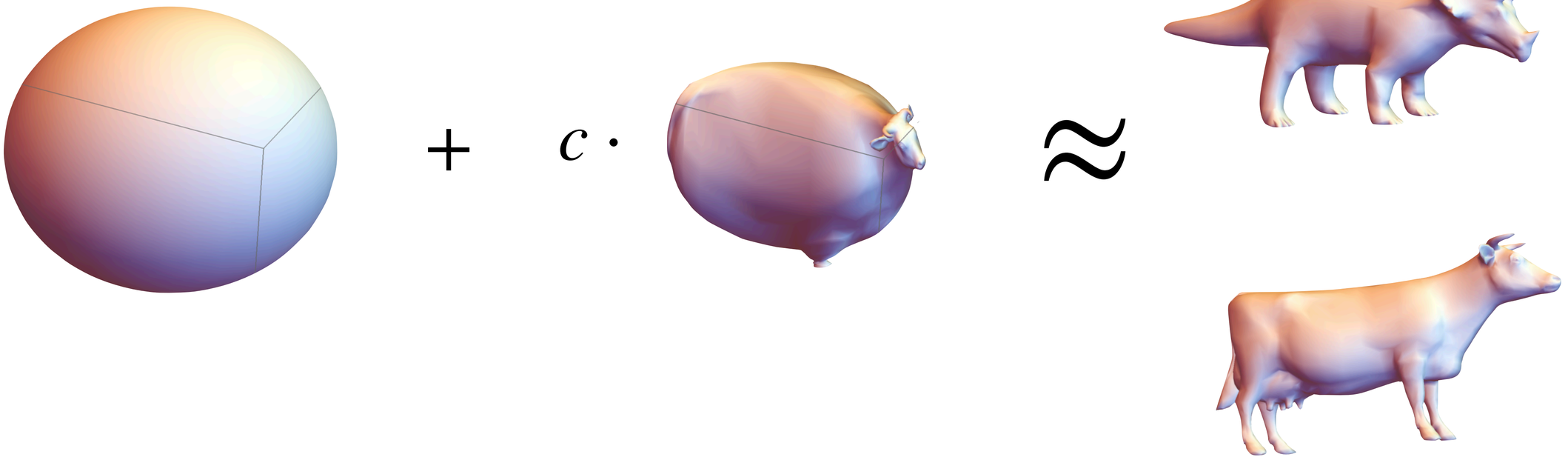
The SMEFT: the way to probe new physics beyond the direct collider energy reach

The SMEFT philosophy

The SM

EFT

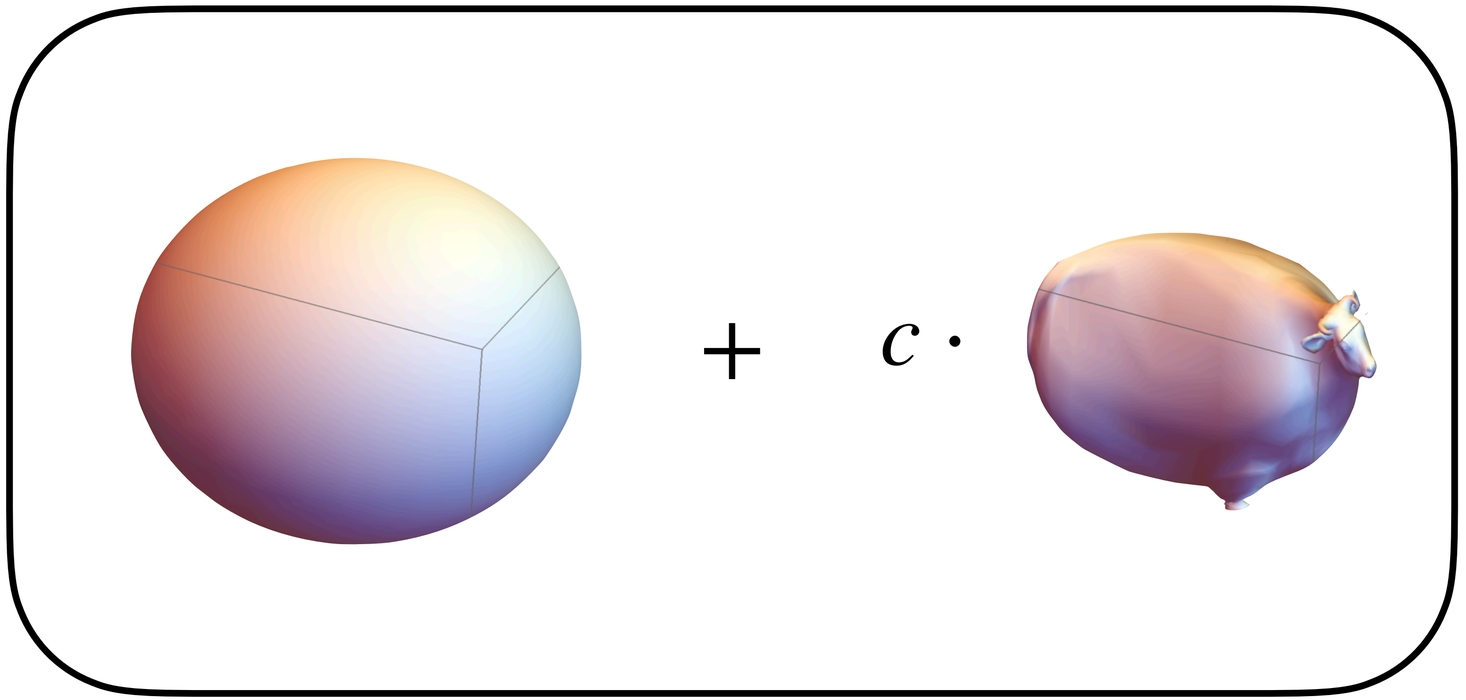
New Physics



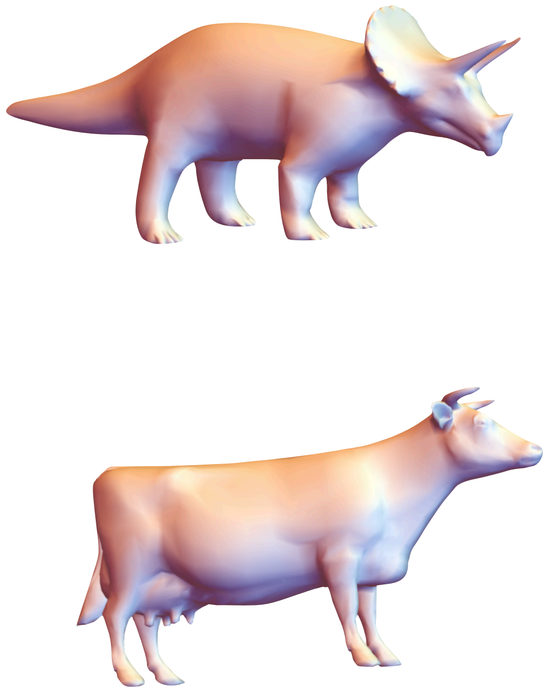
Goal: find the value of c, and precisely!

The SMEFT philosophy

The SM EFT New Physics



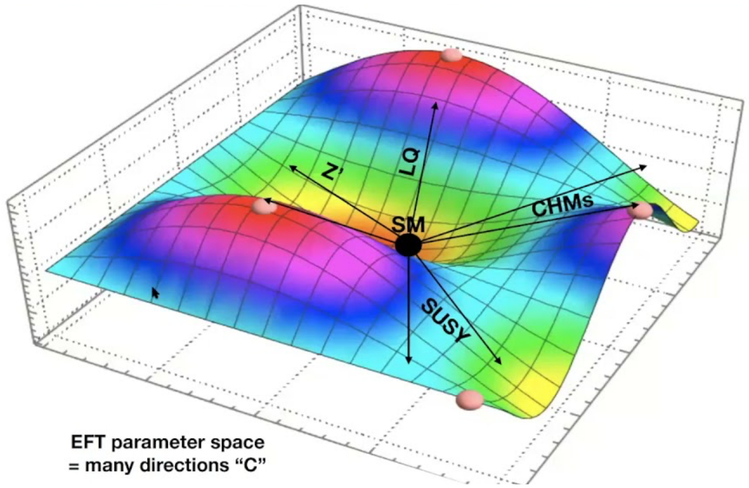
$\approx$



Goal: find the value of c, and precisely!

SMEFT requirements

$$\mathscr{L}_{\text{SMEFT}} = \mathscr{L}_{\text{SM}} + \sum_i^{N_{d5}} \frac{c_i}{\Lambda} \mathcal{O}_i^{(5)} + \sum_i^{N_{d6}} \frac{c_i}{\Lambda^2} \mathcal{O}_i^{(6)} + \ldots$$



- The SMEFT parameterises the **theory space around the SM**
- Defines a perfectly valid QFT that is **renormalisable** order by order in Λ
- Operators are constructed out of **SM fields only** and respects the SM **symmetries**
- SM-like EWSB
- Forms a **complete basis** at any given mass dimension

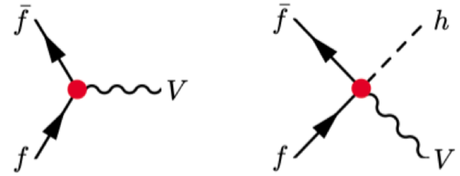
The Warsaw basis

Grzadkowski et al, arXiv: 1008.4884

X^3		φ^6 and $\varphi^4 D^2$		$\psi^2\varphi^3$	
Q_G	$f^{ABC}G^{A\nu}_{\mu}G^{B\rho}_{\nu}G^{C\mu}_{\rho}$	Q_{φ}	$(\varphi^\dagger\varphi)^3$	$Q_{e\varphi}$	$(\varphi^\dagger\varphi)(\bar{l}_p e_r \varphi)$
$Q_{\widetilde{G}}$	$f^{ABC}\widetilde{G}^{A\nu}_{\mu}G^{B\rho}_{\nu}G^{C\mu}_{\rho}$	$Q_{\varphi\Box}$	$(\varphi^\dagger\varphi)\Box(\varphi^\dagger\varphi)$	$Q_{u\varphi}$	$(\varphi^\dagger\varphi)(\bar{q}_p u_r \widetilde{\varphi})$
Q_W	$\varepsilon^{IJK}W^{I\nu}_{\mu}W^{J\rho}_{\nu}W^{K\mu}_{\rho}$	$Q_{\varphi D}$	$\left(\varphi^\dagger D^\mu\varphi\right)^\star\left(\varphi^\dagger D_\mu\varphi\right)$	$Q_{d\varphi}$	$(\varphi^\dagger\varphi)(\bar{q}_p d_r \varphi)$
$Q_{\widetilde{W}}$	$\varepsilon^{IJK}\widetilde{W}^{I\nu}_{\mu}W^{J\rho}_{\nu}W^{K\mu}_{\rho}$				
$X^2\varphi^2$		$\psi^2 X\varphi$		$\psi^2\varphi^2 D$	
$Q_{\varphi G}$	$\varphi^\dagger\varphi\, G^A_{\mu\nu}G^{A\mu\nu}$	Q_{eW}	$(\bar{l}_p\sigma^{\mu\nu}e_r)\tau^I\varphi W^I_{\mu\nu}$	$Q^{(1)}_{\varphi l}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{l}_p\gamma^\mu l_r)$
$Q_{\varphi\widetilde{G}}$	$\varphi^\dagger\varphi\,\widetilde{G}^A_{\mu\nu}G^{A\mu\nu}$	Q_{eB}	$(\bar{l}_p\sigma^{\mu\nu}e_r)\varphi B_{\mu\nu}$	$Q^{(3)}_{\varphi l}$	$(\varphi^\dagger i\overleftrightarrow{D}^I_\mu\varphi)(\bar{l}_p\tau^I\gamma^\mu l_r)$
$Q_{\varphi W}$	$\varphi^\dagger\varphi\, W^I_{\mu\nu}W^{I\mu\nu}$	Q_{uG}	$(\bar{q}_p\sigma^{\mu\nu}T^A u_r)\widetilde{\varphi}\, G^A_{\mu\nu}$	$Q_{\varphi e}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{e}_p\gamma^\mu e_r)$
$Q_{\varphi\widetilde{W}}$	$\varphi^\dagger\varphi\,\widetilde{W}^I_{\mu\nu}W^{I\mu\nu}$	Q_{uW}	$(\bar{q}_p\sigma^{\mu\nu}u_r)\tau^I\widetilde{\varphi}\, W^I_{\mu\nu}$	$Q^{(1)}_{\varphi q}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{q}_p\gamma^\mu q_r)$
$Q_{\varphi B}$	$\varphi^\dagger\varphi\, B_{\mu\nu}B^{\mu\nu}$	Q_{uB}	$(\bar{q}_p\sigma^{\mu\nu}u_r)\widetilde{\varphi}\, B_{\mu\nu}$	$Q^{(3)}_{\varphi q}$	$(\varphi^\dagger i\overleftrightarrow{D}^I_\mu\varphi)(\bar{q}_p\tau^I\gamma^\mu q_r)$
$Q_{\varphi\widetilde{B}}$	$\varphi^\dagger\varphi\,\widetilde{B}_{\mu\nu}B^{\mu\nu}$	Q_{dG}	$(\bar{q}_p\sigma^{\mu\nu}T^A d_r)\varphi\, G^A_{\mu\nu}$	$Q_{\varphi u}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{u}_p\gamma^\mu u_r)$
$Q_{\varphi WB}$	$\varphi^\dagger\tau^I\varphi\, W^I_{\mu\nu}B^{\mu\nu}$	Q_{dW}	$(\bar{q}_p\sigma^{\mu\nu}d_r)\tau^I\varphi\, W^I_{\mu\nu}$	$Q_{\varphi d}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{d}_p\gamma^\mu d_r)$
$Q_{\varphi\widetilde{W}B}$	$\varphi^\dagger\tau^I\varphi\,\widetilde{W}^I_{\mu\nu}B^{\mu\nu}$	Q_{dB}	$(\bar{q}_p\sigma^{\mu\nu}d_r)\varphi\, B_{\mu\nu}$	$Q_{\varphi ud}$	$i(\widetilde{\varphi}^\dagger D_\mu\varphi)(\bar{u}_p\gamma^\mu d_r)$

Adapted from K. Mimasu

Current operators



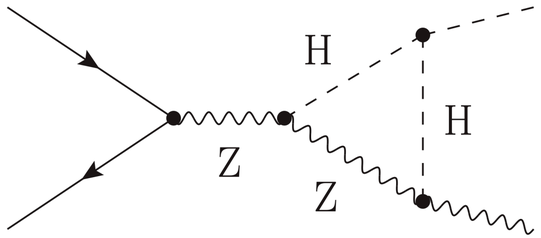
- Shift SM $f\bar{f}V$ couplings
- $f\bar{f}Vh$ contact interactions
- Enters the EWPOs at LEP for example

Contact interactions grow with energy and thus provide a sensitive probe in the TeV region

The Warsaw basis

X^3		φ^6 and $\varphi^4 D^2$		$\psi^2\varphi^3$	
Q_G	$f^{ABC}G_\mu^{A\nu}G_\nu^{B\rho}G_\rho^{C\mu}$	Q_φ	$(\varphi^\dagger\varphi)^3$	$Q_{e\varphi}$	$(\varphi^\dagger\varphi)(\bar{l}_p e_r \varphi)$
$Q_{\widetilde{G}}$	$f^{ABC}\widetilde{G}_\mu^{A\nu}G_\nu^{B\rho}G_\rho^{C\mu}$	$Q_{\varphi\Box}$	$(\varphi^\dagger\varphi)\Box(\varphi^\dagger\varphi)$	$Q_{u\varphi}$	$(\varphi^\dagger\varphi)(\bar{q}_p u_r \widetilde{\varphi})$
Q_W	$\varepsilon^{IJK}W_\mu^{I\nu}W_\nu^{J\rho}W_\rho^{K\mu}$	$Q_{\varphi D}$	$\left(\varphi^\dagger D^\mu\varphi\right)^\star\left(\varphi^\dagger D_\mu\varphi\right)$	$Q_{d\varphi}$	$(\varphi^\dagger\varphi)(\bar{q}_p d_r \varphi)$
$Q_{\widetilde{W}}$	$\varepsilon^{IJK}\widetilde{W}_\mu^{I\nu}W_\nu^{J\rho}W_\rho^{K\mu}$				
$X^2\varphi^2$		$\psi^2 X\varphi$		$\psi^2\varphi^2 D$	
$Q_{\varphi G}$	$\varphi^\dagger\varphi\, G_{\mu\nu}^A G^{A\mu\nu}$	Q_{eW}	$(\bar{l}_p\sigma^{\mu\nu}e_r)\tau^I\varphi W_{\mu\nu}^I$	$Q_{\varphi l}^{(1)}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{l}_p\gamma^\mu l_r)$
$Q_{\varphi\widetilde{G}}$	$\varphi^\dagger\varphi\,\widetilde{G}_{\mu\nu}^A G^{A\mu\nu}$	Q_{eB}	$(\bar{l}_p\sigma^{\mu\nu}e_r)\varphi B_{\mu\nu}$	$Q_{\varphi l}^{(3)}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu^I\varphi)(\bar{l}_p\tau^I\gamma^\mu l_r)$
$Q_{\varphi W}$	$\varphi^\dagger\varphi\, W_{\mu\nu}^I W^{I\mu\nu}$	Q_{uG}	$(\bar{q}_p\sigma^{\mu\nu}T^A u_r)\widetilde{\varphi}\, G_{\mu\nu}^A$	$Q_{\varphi e}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{e}_p\gamma^\mu e_r)$
$Q_{\varphi\widetilde{W}}$	$\varphi^\dagger\varphi\,\widetilde{W}_{\mu\nu}^I W^{I\mu\nu}$	Q_{uW}	$(\bar{q}_p\sigma^{\mu\nu}u_r)\tau^I\widetilde{\varphi}\, W_{\mu\nu}^I$	$Q_{\varphi q}^{(1)}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{q}_p\gamma^\mu q_r)$
$Q_{\varphi B}$	$\varphi^\dagger\varphi\, B_{\mu\nu}B^{\mu\nu}$	Q_{uB}	$(\bar{q}_p\sigma^{\mu\nu}u_r)\widetilde{\varphi}\, B_{\mu\nu}$	$Q_{\varphi q}^{(3)}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu^I\varphi)(\bar{q}_p\tau^I\gamma^\mu q_r)$
$Q_{\varphi\widetilde{B}}$	$\varphi^\dagger\varphi\,\widetilde{B}_{\mu\nu}B^{\mu\nu}$	Q_{dG}	$(\bar{q}_p\sigma^{\mu\nu}T^A d_r)\varphi\, G_{\mu\nu}^A$	$Q_{\varphi u}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{u}_p\gamma^\mu u_r)$
$Q_{\varphi WB}$	$\varphi^\dagger\tau^I\varphi\, W_{\mu\nu}^I B^{\mu\nu}$	Q_{dW}	$(\bar{q}_p\sigma^{\mu\nu}d_r)\tau^I\varphi\, W_{\mu\nu}^I$	$Q_{\varphi d}$	$(\varphi^\dagger i\overleftrightarrow{D}_\mu\varphi)(\bar{d}_p\gamma^\mu d_r)$
$Q_{\varphi\widetilde{W}B}$	$\varphi^\dagger\tau^I\varphi\,\widetilde{W}_{\mu\nu}^I B^{\mu\nu}$	Q_{dB}	$(\bar{q}_p\sigma^{\mu\nu}d_r)\varphi\, B_{\mu\nu}$	$Q_{\varphi ud}$	$i(\widetilde{\varphi}^\dagger D_\mu\varphi)(\bar{u}_p\gamma^\mu d_r)$

Bosonic operators



- φ^6 modifies the Higgs potential and its self-coupling κ_3
- Enters at NLO EW in $e^+e^- \to ZH$
- Provides an indirect probe to κ_3 even when HH production is kinematically not accessible!

Asteriadis et al, arXiv: 2406.03557

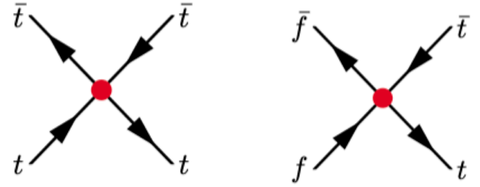
The Warsaw basis

X^3	φ^6 and $\varphi^4 D^2$	$\psi^2\varphi^3$

	$(\bar{L}L)(\bar{L}L)$		$(\bar{R}R)(\bar{R}R)$		$(\bar{L}L)(\bar{R}R)$	
Q_G	Q_{ll}	$(\bar{l}_p\gamma_\mu l_r)(\bar{l}_s\gamma^\mu l_t)$	Q_{ee}	$(\bar{e}_p\gamma_\mu e_r)(\bar{e}_s\gamma^\mu e_t)$	Q_{le}	$(\bar{l}_p\gamma_\mu l_r)(\bar{e}_s\gamma^\mu e_t)$
$Q_{\widetilde{G}}$	$Q_{qq}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{q}_s\gamma^\mu q_t)$	Q_{uu}	$(\bar{u}_p\gamma_\mu u_r)(\bar{u}_s\gamma^\mu u_t)$	Q_{lu}	$(\bar{l}_p\gamma_\mu l_r)(\bar{u}_s\gamma^\mu u_t)$
Q_W	$Q_{qq}^{(3)}$	$(\bar{q}_p\gamma_\mu\tau^I q_r)(\bar{q}_s\gamma^\mu\tau^I q_t)$	Q_{dd}	$(\bar{d}_p\gamma_\mu d_r)(\bar{d}_s\gamma^\mu d_t)$	Q_{ld}	$(\bar{l}_p\gamma_\mu l_r)(\bar{d}_s\gamma^\mu d_t)$
$Q_{\widetilde{W}}$	$Q_{lq}^{(1)}$	$(\bar{l}_p\gamma_\mu l_r)(\bar{q}_s\gamma^\mu q_t)$	Q_{eu}	$(\bar{e}_p\gamma_\mu e_r)(\bar{u}_s\gamma^\mu u_t)$	Q_{qe}	$(\bar{q}_p\gamma_\mu q_r)(\bar{e}_s\gamma^\mu e_t)$
	$Q_{lq}^{(3)}$	$(\bar{l}_p\gamma_\mu\tau^I l_r)(\bar{q}_s\gamma^\mu\tau^I q_t)$	Q_{ed}	$(\bar{e}_p\gamma_\mu e_r)(\bar{d}_s\gamma^\mu d_t)$	$Q_{qu}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{u}_s\gamma^\mu u_t)$
$Q_{\varphi G}$			$Q_{ud}^{(1)}$	$(\bar{u}_p\gamma_\mu u_r)(\bar{d}_s\gamma^\mu d_t)$	$Q_{qu}^{(8)}$	$(\bar{q}_p\gamma_\mu T^A q_r)(\bar{u}_s\gamma^\mu T^A u_t)$
$Q_{\varphi\widetilde{G}}$			$Q_{ud}^{(8)}$	$(\bar{u}_p\gamma_\mu T^A u_r)(\bar{d}_s\gamma^\mu T^A d_t)$	$Q_{qd}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{d}_s\gamma^\mu d_t)$
$Q_{\varphi W}$					$Q_{qd}^{(8)}$	$(\bar{q}_p\gamma_\mu T^A q_r)(\bar{d}_s\gamma^\mu T^A d_t)$

$Q_{\varphi\widetilde{W}}$	$(\bar{L}R)(\bar{R}L)$ and $(\bar{L}R)(\bar{L}R)$		B-violating	
$Q_{\varphi B}$	Q_{ledq}	$(\bar{l}_p^j e_r)(\bar{d}_s q_t^j)$	Q_{duq}	$\varepsilon^{\alpha\beta\gamma}\varepsilon_{jk}\left[(d_p^\alpha)^T C u_r^\beta\right]\left[(q_s^{\gamma j})^T C l_t^k\right]$
$Q_{\varphi\widetilde{B}}$	$Q_{quqd}^{(1)}$	$(\bar{q}_p^j u_r)\varepsilon_{jk}(\bar{q}_s^k d_t)$	Q_{qqu}	$\varepsilon^{\alpha\beta\gamma}\varepsilon_{jk}\left[(q_p^{\alpha j})^T C q_r^{\beta k}\right]\left[(u_s^\gamma)^T C e_t\right]$
$Q_{\varphi WB}$	$Q_{quqd}^{(8)}$	$(\bar{q}_p^j T^A u_r)\varepsilon_{jk}(\bar{q}_s^k T^A d_t)$	Q_{qqq}	$\varepsilon^{\alpha\beta\gamma}\varepsilon_{jn}\varepsilon_{km}\left[(q_p^{\alpha j})^T C q_r^{\beta k}\right]\left[(q_s^{\gamma m})^T C l_t^n\right]$
$Q_{\varphi\widetilde{W}B}$	$Q_{lequ}^{(1)}$	$(\bar{l}_p^j e_r)\varepsilon_{jk}(\bar{q}_s^k u_t)$	Q_{duu}	$\varepsilon^{\alpha\beta\gamma}\left[(d_p^\alpha)^T C u_r^\beta\right]\left[(u_s^\gamma)^T C e_t\right]$
	$Q_{lequ}^{(3)}$	$(\bar{l}_p^j\sigma_{\mu\nu} e_r)\varepsilon_{jk}(\bar{q}_s^k\sigma^{\mu\nu} u_t)$		

Four-fermion operators



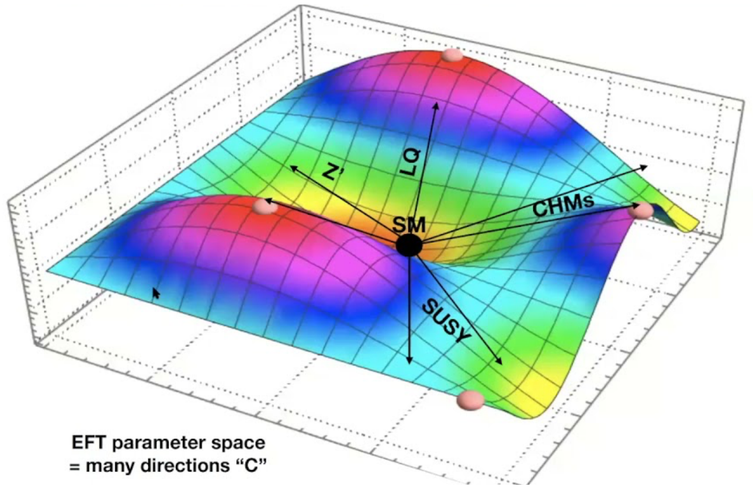
- Contact interactions
- 2205 B/L preserving

All in all, we have 2499 operators at dim-6!

Often reduced by flavour symmetries

SMEFT predictions

$$\mathscr{L}_{\text{SMEFT}} = \mathscr{L}_{\text{SM}} + \sum_i^{N_{d5}} \frac{c_i}{\Lambda} \mathcal{O}_i^{(5)} + \sum_i^{N_{d6}} \frac{c_i}{\Lambda^2} \mathcal{O}_i^{(6)} + \ldots$$



$$\sigma_{\text{SMEFT}}(c, \Lambda) = \sigma_{\text{SM}} \times \left(1 + \sum_i^{N_{d6}} \kappa_i \frac{c_i}{\Lambda^2} + \sum_{i<j}^{N_{d6}} \tilde{\kappa}_{ij} \frac{c_i \cdot c_j}{\Lambda^4} + \mathcal{O}\left(\Lambda^{-6}\right) \right)$$

State of the art SM predictions

Linear EFT corrections: interference with SM, computed with SMEFT@NLO

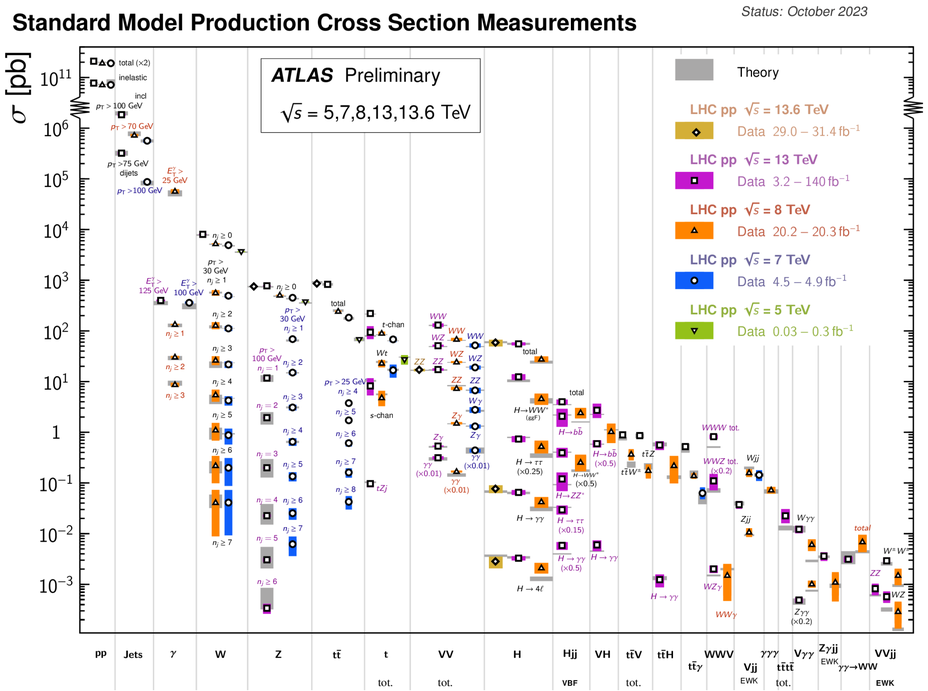
Quadratic EFT corrections, computed with SMEFT@NLO

Degrande et al, arXiv: 2008.11743

Building the likelihood

$$\chi^2 = \frac{1}{n_{\text{dat}}} \sum_{i,j=1}^{n_{\text{dat}}} \left(\sigma_{i,\text{SMEFT}}(c) - \sigma_{i,\text{exp}}\right) \left(\text{cov}^{-1}\right)_{ij} \left(\sigma_{j,\text{SMEFT}}(c) - \sigma_{j,\text{exp}}\right)$$

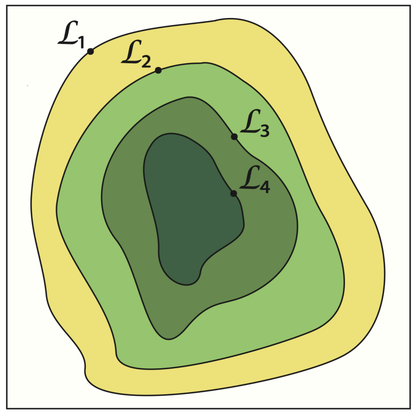
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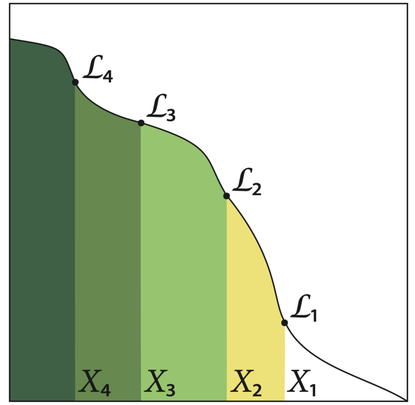


$$\text{cov}^{(\text{tot})}_{ij} = \text{cov}^{(\text{th})}_{ij} + \text{cov}^{(\text{exp})}_{ij}$$

Analytic if $\mathcal{O}\left(\Lambda^{-2}\right)$, fast!

Nested Sampling $\mathcal{O}\left(\Lambda^{-2}\right)$ or $\mathcal{O}\left(\Lambda^{-4}\right)$

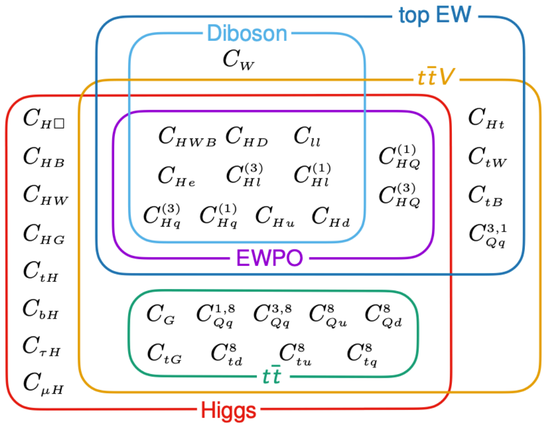




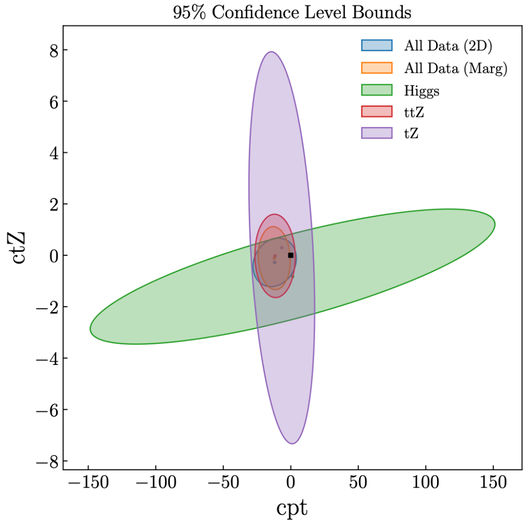
Feroz et al [1306.2144]

Why global SMEFT fits?

- **Cross-talk** between Higgs, top, diboson and EWPO (and flavour and low energy observables) requires a global analysis
- **Challenge**: a large number of operators, with many datasets needed to break degeneracies

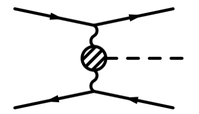


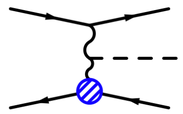
[2012.02779] Fitmaker collaboration



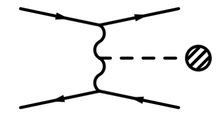
[2105.00006] SMEFiT collaboration

One observable can be influenced by many operators

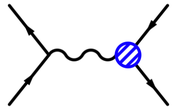




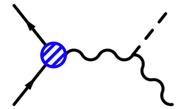
Higgs decay



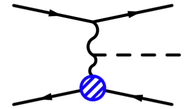
One operator can contribute to many different observables



$e^+e^- \to f\bar{f}$



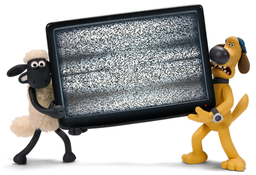
Zh production



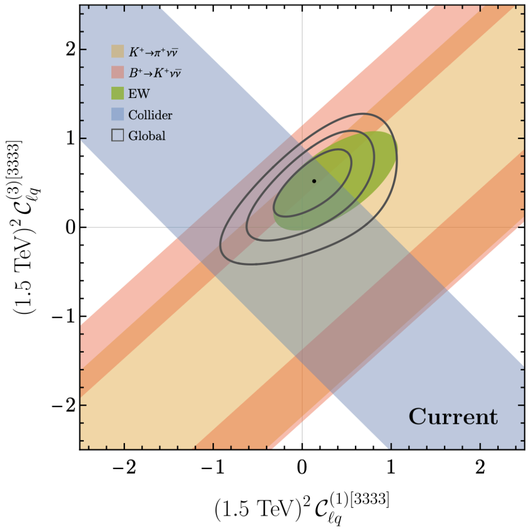
Weak boson fusion
Higgs production

Anke Biekötter - HET seminar Brookhaven

Previously on global fits…



- **SMEFiT**: EW + Higgs + diboson + top + projections, NLO, quadratic - *[2105.00006, 2309.04523 , 2404.12809]*
- **ATLAS**: EW + Higgs, LO, quadratic - *[ATL-PHYS-PUB-2022-037]*
- **simuNET:** simultaneous EFT + PDF fit in EW + Higgs + diboson + top, NLO, linear - *[2402.03308]*
- **Fitmaker**: EW + Higgs + top + diboson, linear - *[2012.02779, 2204.05260]*
- **SFitter**: EW + Higgs, top, NLO, quadratic - *[1812.07587, 1910.03606]*
- **HEPfit:** EW, flavour, projections, LO, linear - *[1910.14012]*
- **TopFitter:** top, linear, LO - *[1901.03164]*
- **EFTfitter:** top + DY + flavour, LO, quadratic, RG effects - *[1605.05585, 2304.12837]*
- **Mainz group:** EW + Higgs + top + flavour + dijet + PV + lepton scattering, NLO, linear - *[2311.04963, 2412.09674]*
- **Zurich group:** EW + flavour + (DY, LEPII, Jet observables), individual, RG effects - *[2311.00020]*
- ……

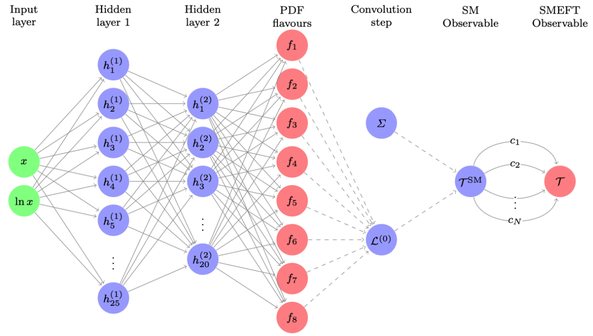


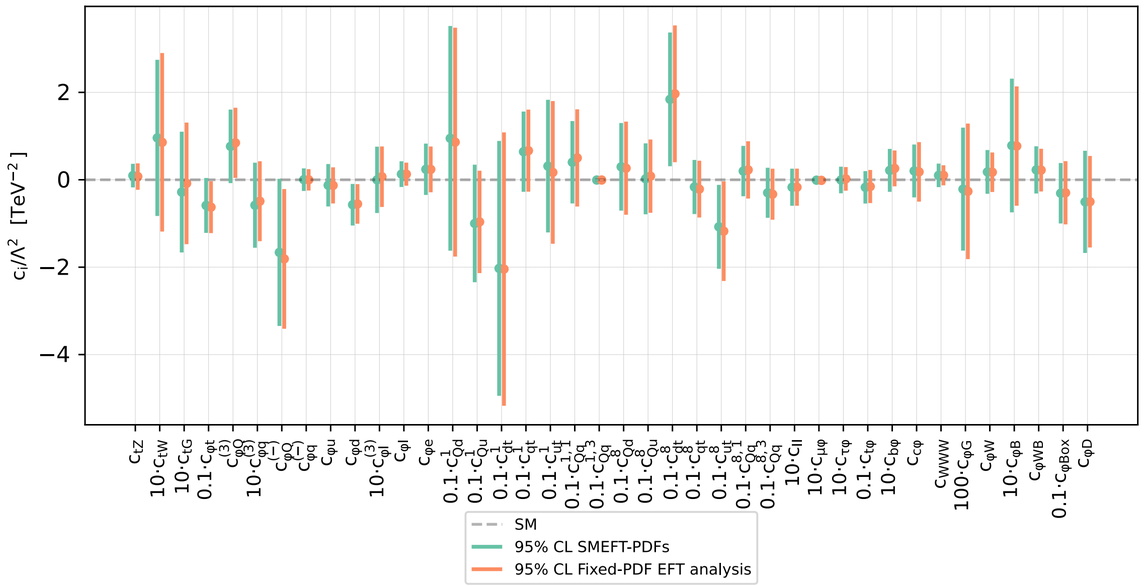
arXiv:2311.00020

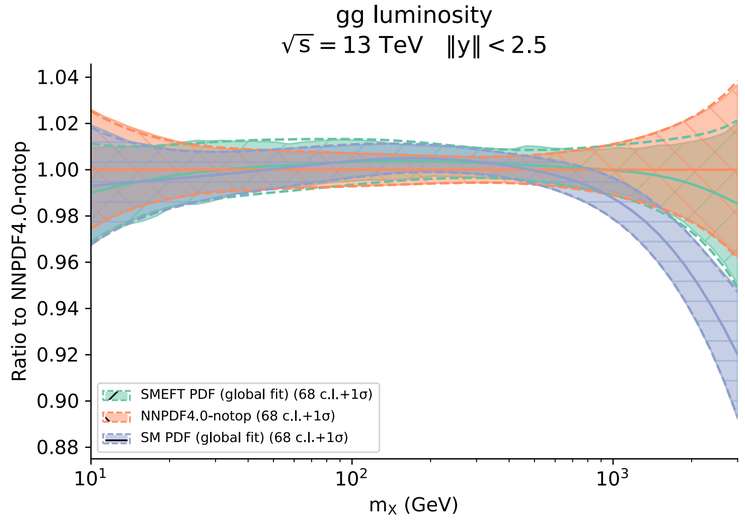
simuNET: a simultaneous PDF + EFT fit

Costantini, Hammou, Madigan, Mantani, Moore, Morales, Ubiali [2402.03308]

- Most EFT global fits assume a fixed PDF set. Ideally, a full treatment fits the EFT and PDF parameters **simultaneously**, as done by simuNET for example
- EFT parameters are stable, while the PDF fits undergo **shifts** at high invariant mass in e.g. the gluon-gluon luminosity







SMEFiT

lhcfitnikhef.github.io/smefit_release

Theory

SM: (N)NLO QCD + NLO EW

EFT: NLO QCD, linear and quadratics, with SMEFT@NLO

NNPDF4.0 no top

Data

447 measurements from **Higgs, top, diboson** and **EWPO**

Full experimental **correlations**

Projections

HL-LHC, FCCee, CEPC

+ Automatic projection module

SMEFiT

dataset_X.json

```
{
  "best_sm": [3.0],
  "scales": [91.0],
  "theory_cov": [[1.0]],
  "LO": {"SM":  1.0, "Op1":  -0.2, "Op1*Op1":  0.4},
  "NLO_QCD": {"SM":  1.5, "Op1":  -0.3, "Op1*Op1":  0.6}
}
```

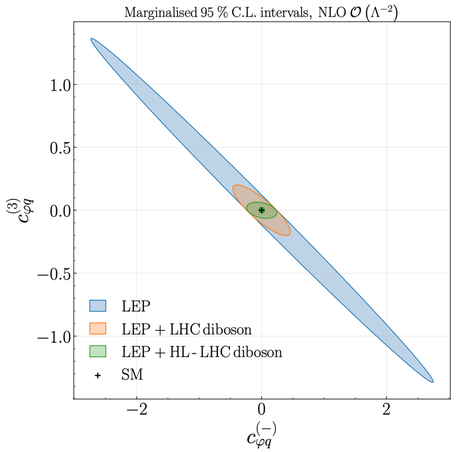
dataset_X.yaml

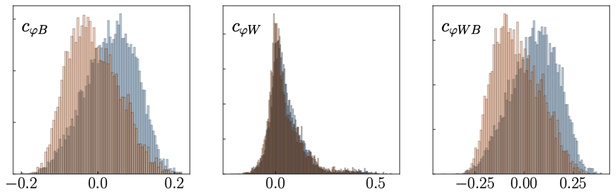
```
dataset_name: ATLAS_ttW_13TeV_2016
doi: 10.1103/PhysRevD.99.072009
location: Figure 13 arxiv preprint
arxiv: 1901.03584
hepdata: https://www.hepdata.net/record/ins1713423
units: fb
description: inclusive ttW cross-section
luminosity: 36.1
num_data: 1
num_sys: 1
data_central: 3.1
statistical_error: 0.1
systematics:
  - 0.2
sys_names: UNCORR
sys_type: MULT
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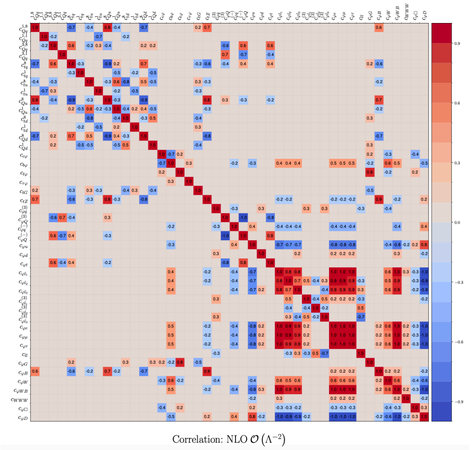
Analysis tools

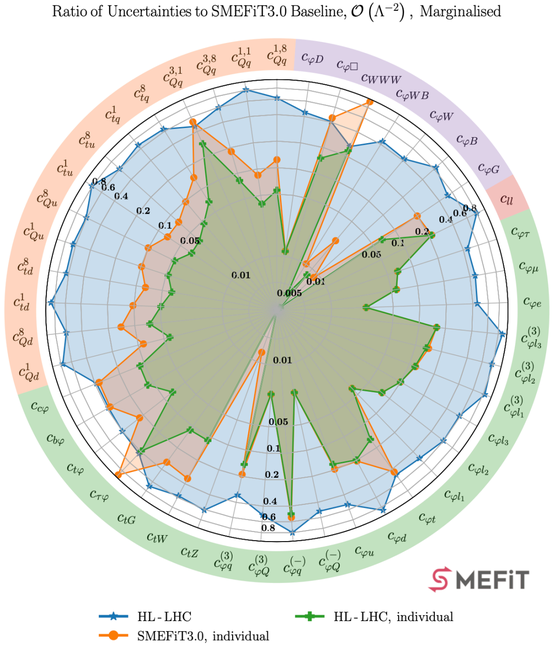
Report module

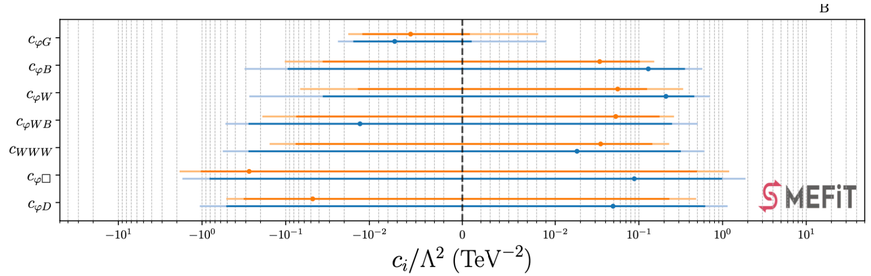
Automatised fit reports that analyse the SMEFiT results

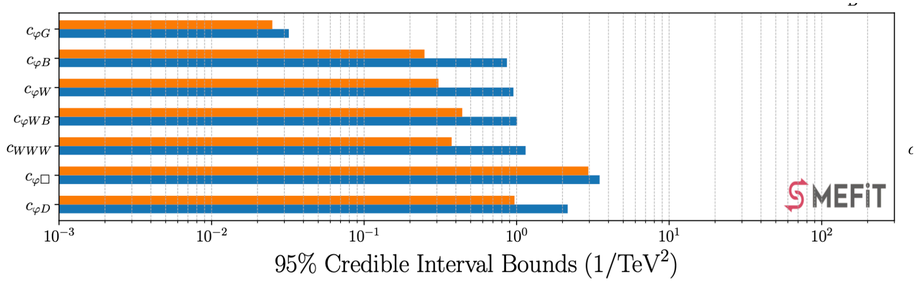










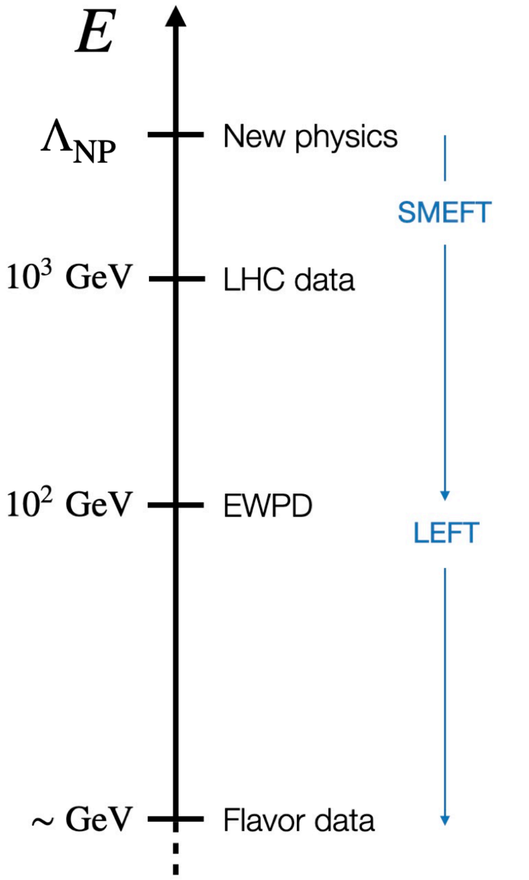


RGE in the SMEFT

Scale differences

- Experimental input to global fits spans **a wide range of energy scales,** from m_Z at LEP to $m_{t\bar{t}} \sim 3\,\text{TeV}$ in tails at LHC
- Scale differences become even more marked when including **flavour data** $Q \sim m_b$
- as wells as sensitivity projections at the FCC-ee around the Z pole

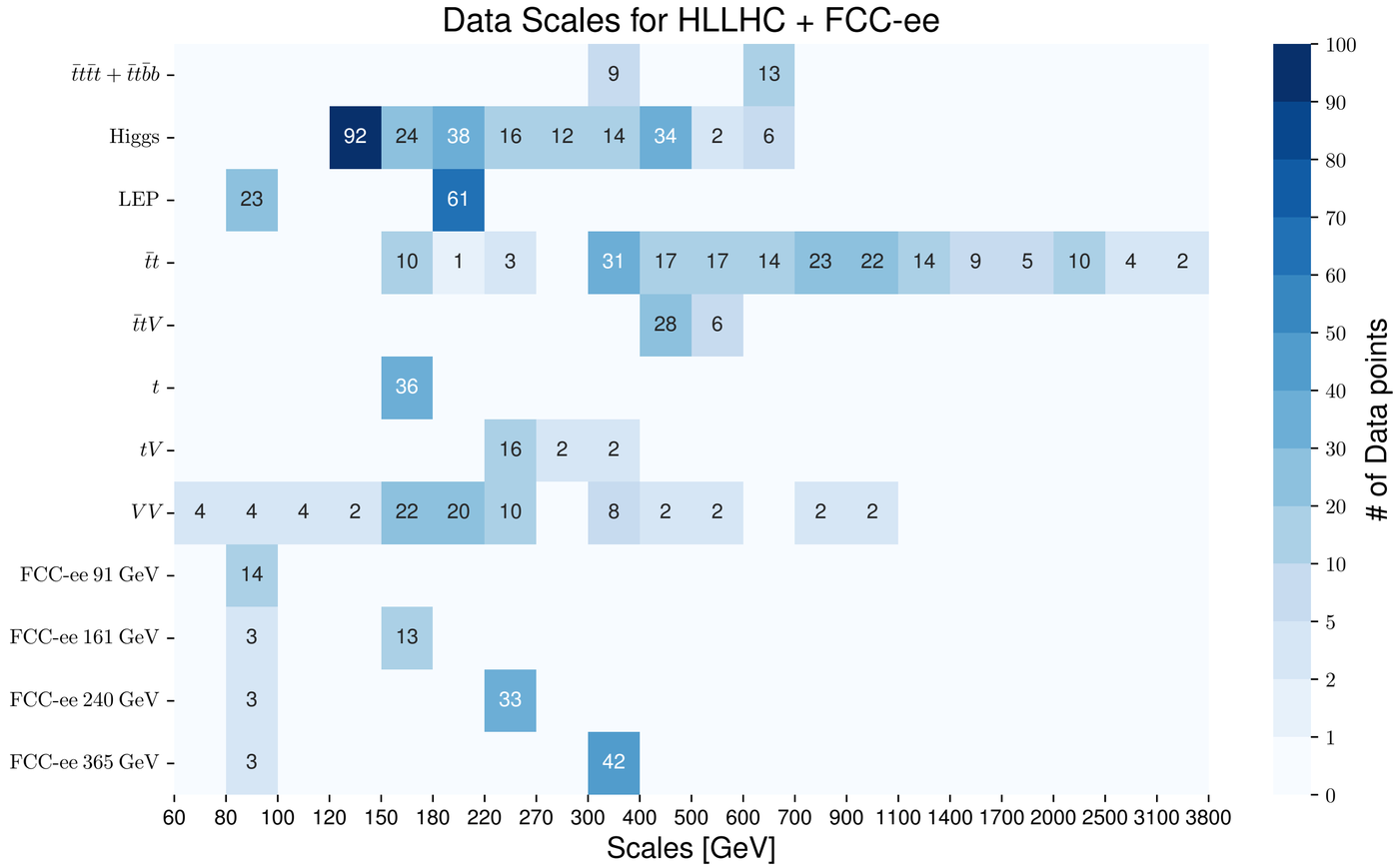
Problem: assuming all measurements entering a global SMEFT fit to correspond to the same energy scale is in general not well justified



B. Stefanek, 3rd ECFA workshop

Scale differences

Problem: assuming that all measurements in global SMEFT fit correspond to the same energy scale is in general not well justified



Mantani, Rojo, Rossia, Vryonidou, ***JtH,*** *arXiv: 2502.20453*

Scale differences

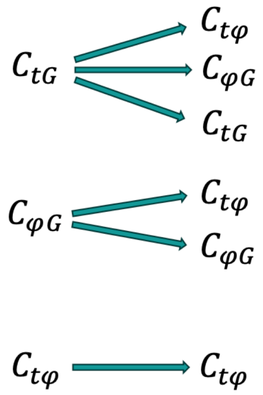
Solution: promote the Wilson coefficient to carry scale dependence

$$\sigma_{\text{SMEFT}}(c(\mu), \Lambda) = \sigma_{\text{SM}} \times \left(1 + \sum_{i}^{N_{d6}} \kappa_i \frac{c_i(\mu)}{\Lambda^2} + \sum_{i<j}^{N_{d6}} \tilde{\kappa}_{ij} \frac{c_i(\mu) \cdot c_j(\mu)}{\Lambda^4} \right)$$

- Wilson coefficients run and mix with energy through the anomalous dimension

$$\frac{dc_i(\mu)}{d\ln\mu} = \sum_{j=1}^{n_{\text{op}}} \gamma_{ij}^{(6)}(\bar{g})\, c_j(\mu)$$

Jenkins, Manohar, Trott, Alonso, arXiv:1308.2627, 1310.4838, 1312.2014



- Implemented in several codes

wilson

[1804.05033] Aebischer et al

DsixTools

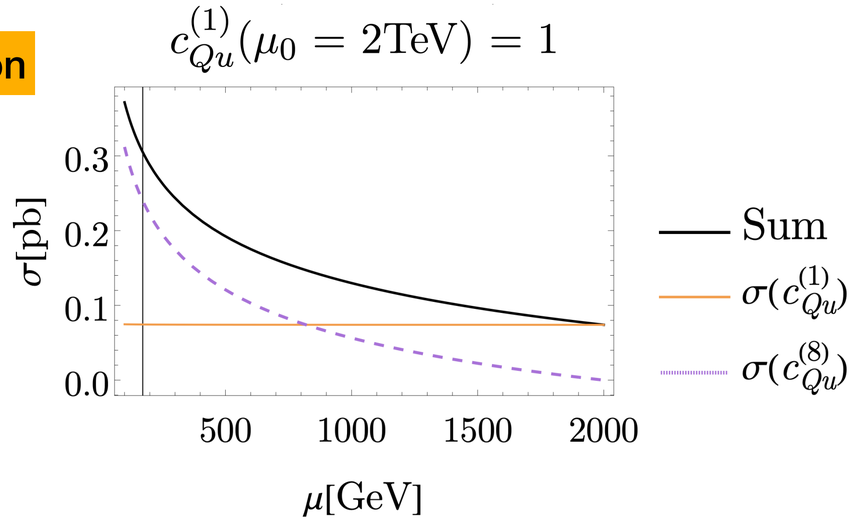
[2010.16341] Fuentes-Martin et al

RGESolver

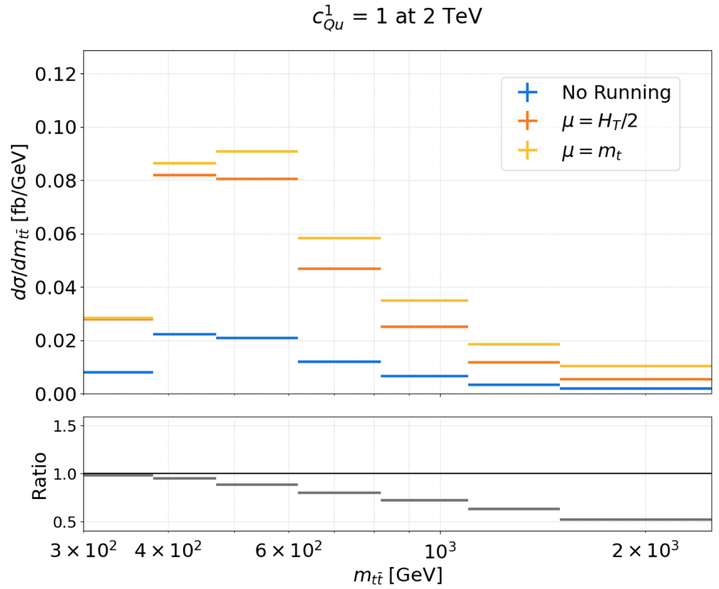
[2210.06838] Di Noi et al

RGE in SMEFT phenomenology

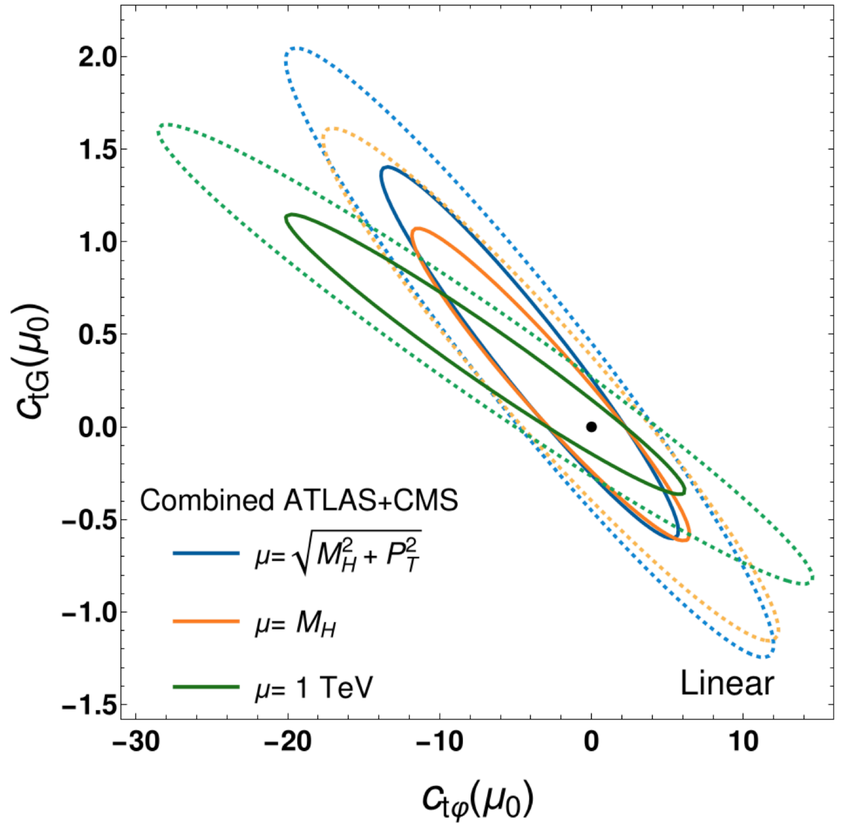
Top pair production



Aoude et al.
arXiv: 2212.05067



Higgs production



Maltoni, Ventura, Vryonidou
arXiv: 2406.06670

RGE in SMEFiT

- We include RG effects in the **Matrix Evolution Approximation** interfaced to Wilson

$$c_i(\mu) = \sum_{j=1}^{n_{\rm op}} \Gamma_{ij}(\mu, \mu_0; \alpha_s, \alpha) c_j(\mu_0)$$

wilson

[1804.05033] Aebischer, Kumar, Straub

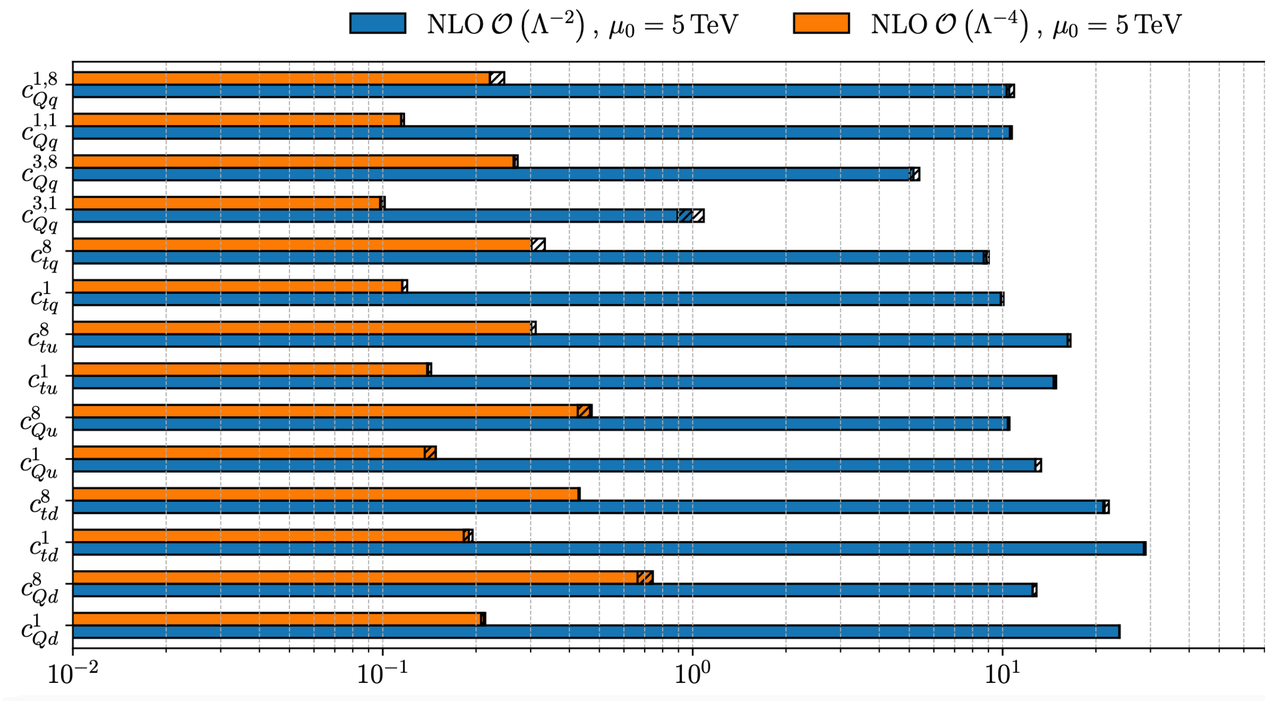
- to express the theory predictions at a common scale μ_0 = 5 TeV

$$\begin{aligned} T_{\rm EFT}(\boldsymbol{c}(\mu)/\Lambda^2) &= T_{\rm SM} + \sum_{i=1}^{n_{\rm op}} \kappa_i \frac{c_i(\mu)}{\Lambda^2} + \sum_{i,j=1}^{n_{\rm op}} \widetilde{\kappa}_{ij} \frac{c_i(\mu) c_j(\mu)}{\Lambda^4} \\ &= T_{\rm SM} + \sum_{i,j=1}^{n_{\rm op}} \kappa_i \Gamma_{ij} \frac{c_j(\mu_0)}{\Lambda^2} + \sum_{i,j,k,\ell=1}^{n_{\rm op}} \widetilde{\kappa}_{ij} \Gamma_{ik} \Gamma_{j\ell} \frac{c_k(\mu_0) c_\ell(\mu_0)}{\Lambda^4} \\ &= T_{\rm SM} + \sum_{j=1}^{n_{\rm op}} \kappa'_j \frac{c_j(\mu_0)}{\Lambda^2} + \sum_{k,\ell=1}^{n_{\rm op}} \widetilde{\kappa}'_{k\ell} \frac{c_k(\mu_0) c_\ell(\mu_0)}{\Lambda^4} \end{aligned}$$

RGE in SMEFiT

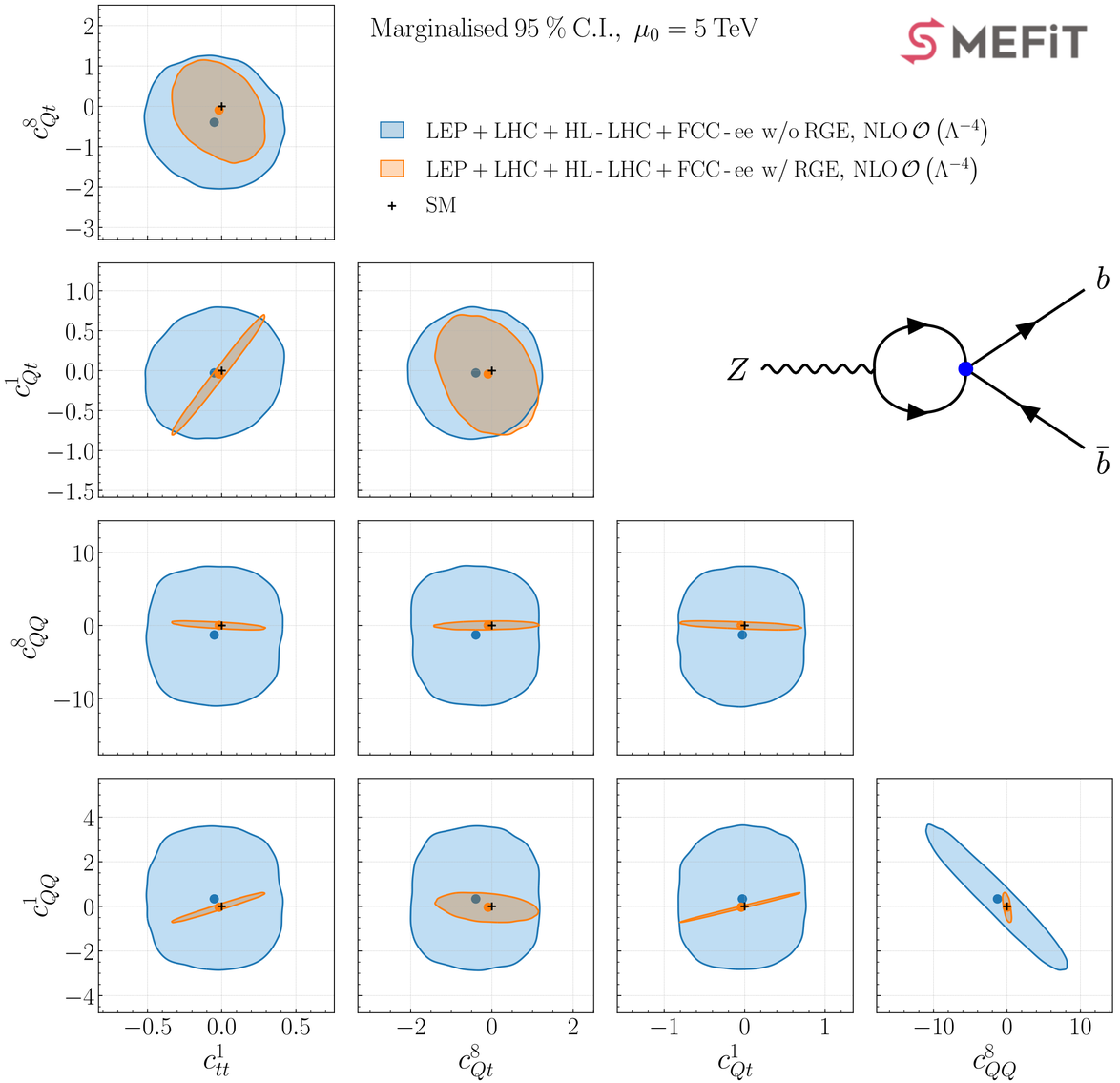
- We associate a fixed characteristic scale choice μ to **each bin**

Scale variations $\mu \to \tilde{\mu} = \kappa\mu$



Process	**Scale Choice μ**	**Process**	**Scale Choice μ**
Higgs (ggF)	$\sqrt{m_H^2 + (p_T^H)^2}$	$t\bar{t}b\bar{b}$	$2m_t$
Higgs (VBF)	$\sqrt{m_H^2 + (p_T^H)^2}$	$t\bar{t}V$	$\sqrt{(2m_t + m_V)^2 + (p_T^V)^2}$
VH	$\sqrt{(m_V + m_H)^2 + (p_T^V)^2}$	tV	$m_t + m_V$ or $\sqrt{(m_t + m_V)^2 + (p_T^t)^2}$
$t\bar{t}H$	$\sqrt{(2m_t + m_H)^2 + (p_T^H)^2}$	W-helicities	m_t
tH	$m_t + m_H$	WZ	m_T^{WZ} or $\sqrt{(m_Z + m_W)^2 + (p_T^Z)^2}$
$t\bar{t}$	m_{tt}	WW	$m_{e\mu}$
Single-t	m_t	V pole (incl. EWPOs)	m_V
$t\bar{t}\gamma$	$2m_t$	Bhabha scattering	$\sqrt{s}$
$t\bar{t}t\bar{t}$	$4m_t$	$e^+e^- \to WW\,/\,t\bar{t}\,/\,f\bar{f}$	$\sqrt{s}$
HH	$2m_H$	$e^+e^- \to ZH$	$\sqrt{s}$

RG in restricted operator basis

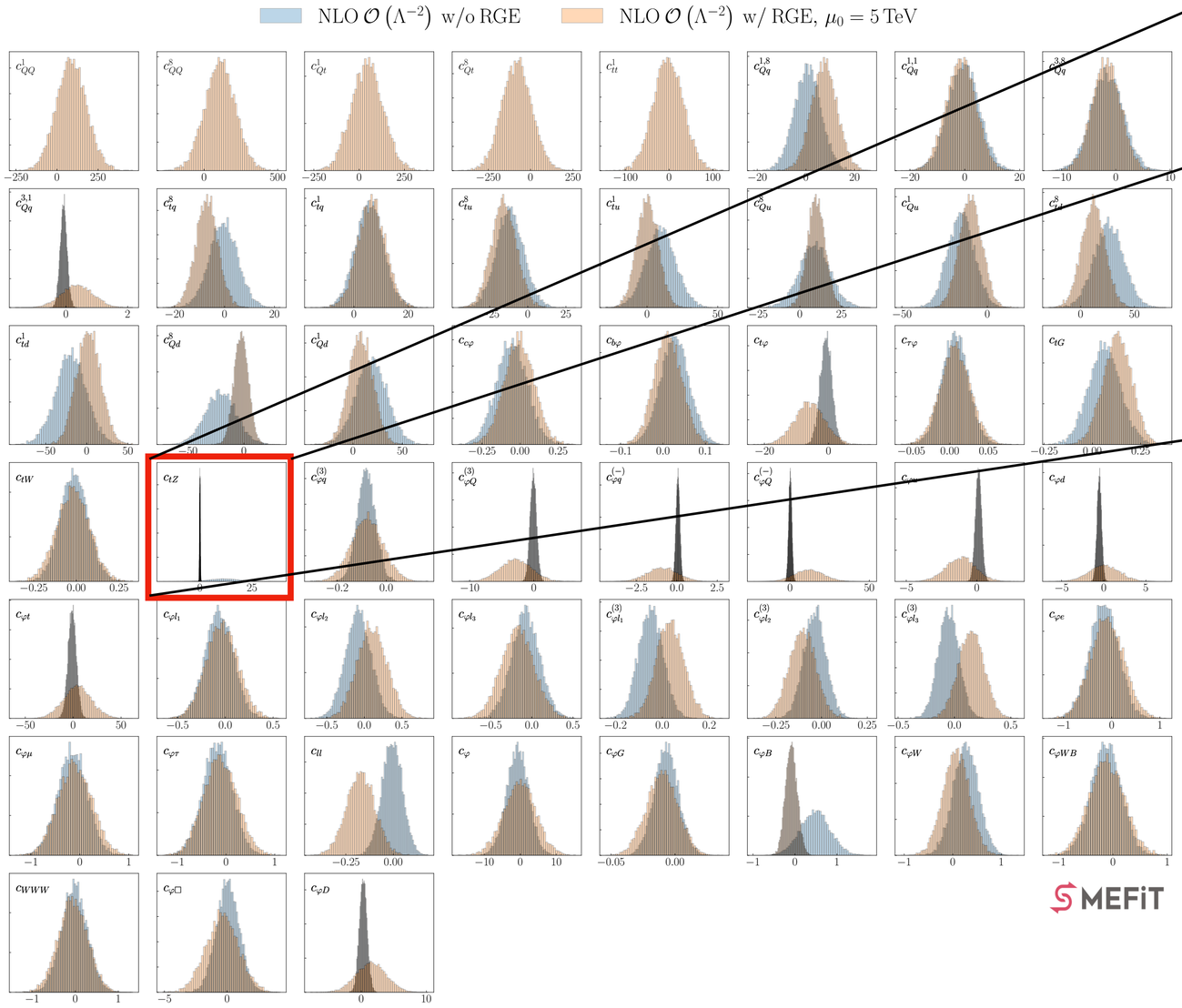


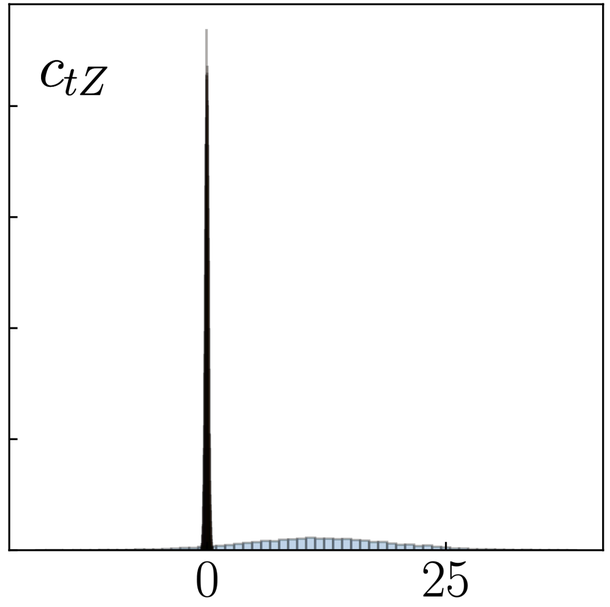
Example: 4 heavy operators flow into operators sensitive to the EWPOs at low energy

In general one may see two competing effects:

- Ill constrained operators may flow into a precisely determined observable
- More operators enter the same observable, making bounds weaker

RG in the global fit



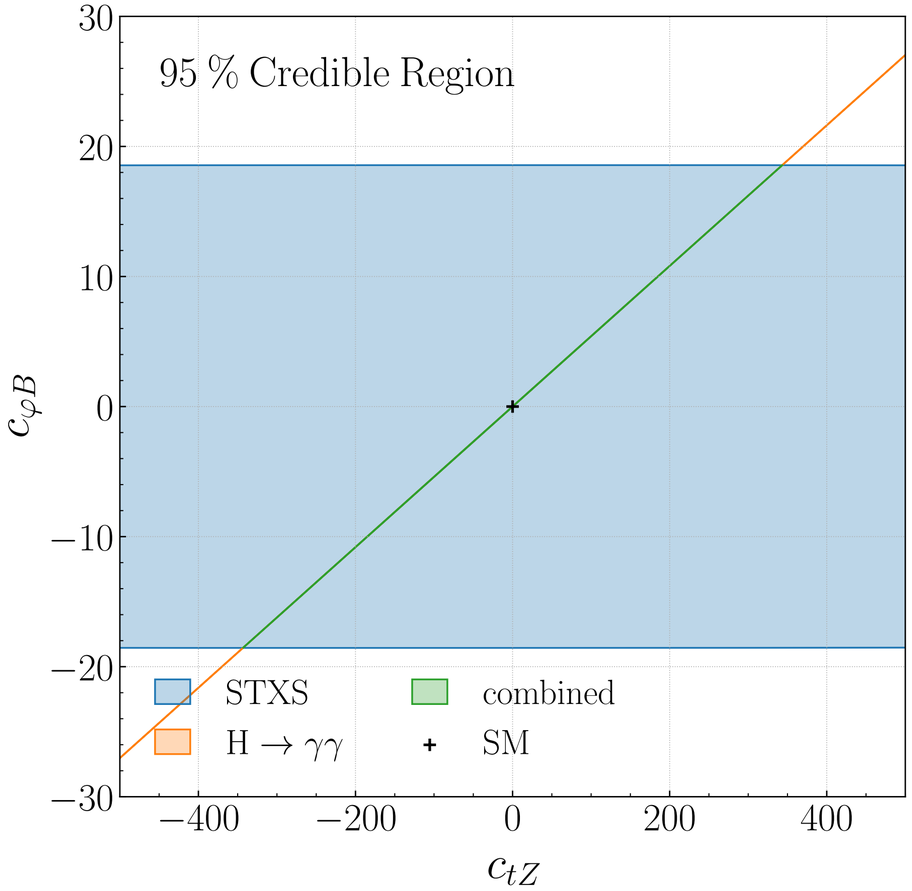


NLO $\mathcal{O}\left(\Lambda^{-2}\right)$ w/o RGE

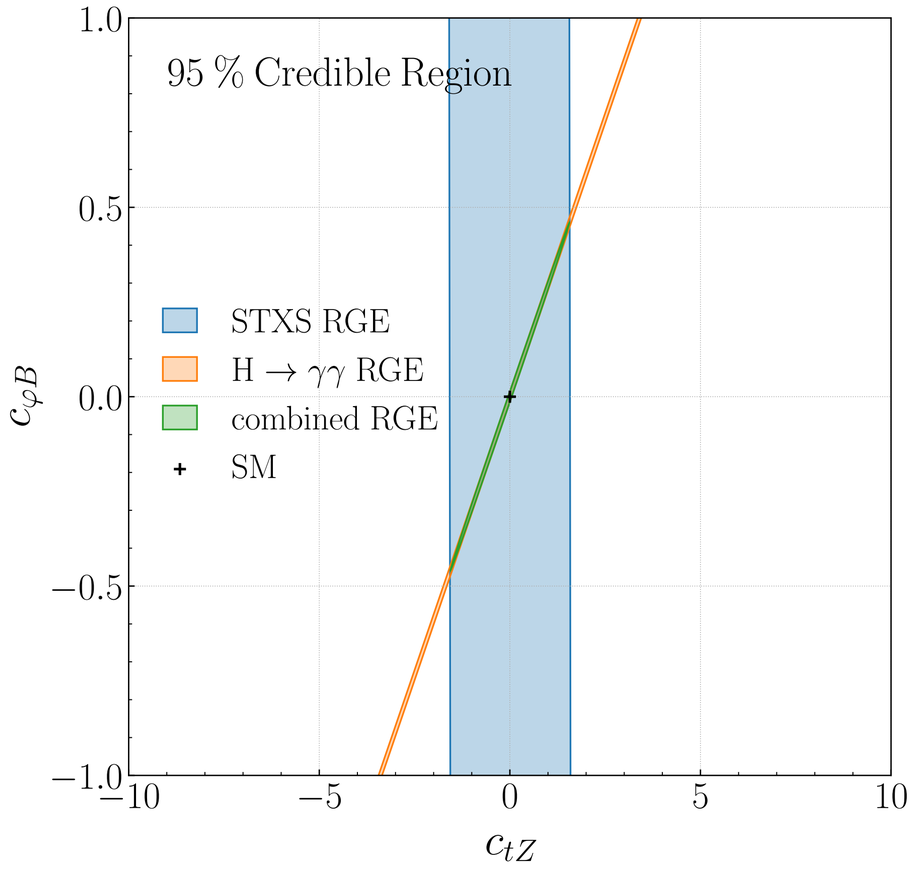
NLO $\mathcal{O}\left(\Lambda^{-2}\right)$ w/ RGE, $\mu_0 = 5\,\text{TeV}$

RG in the global fit

Global fits neglecting RGE effects can severely underestimate the bounds!



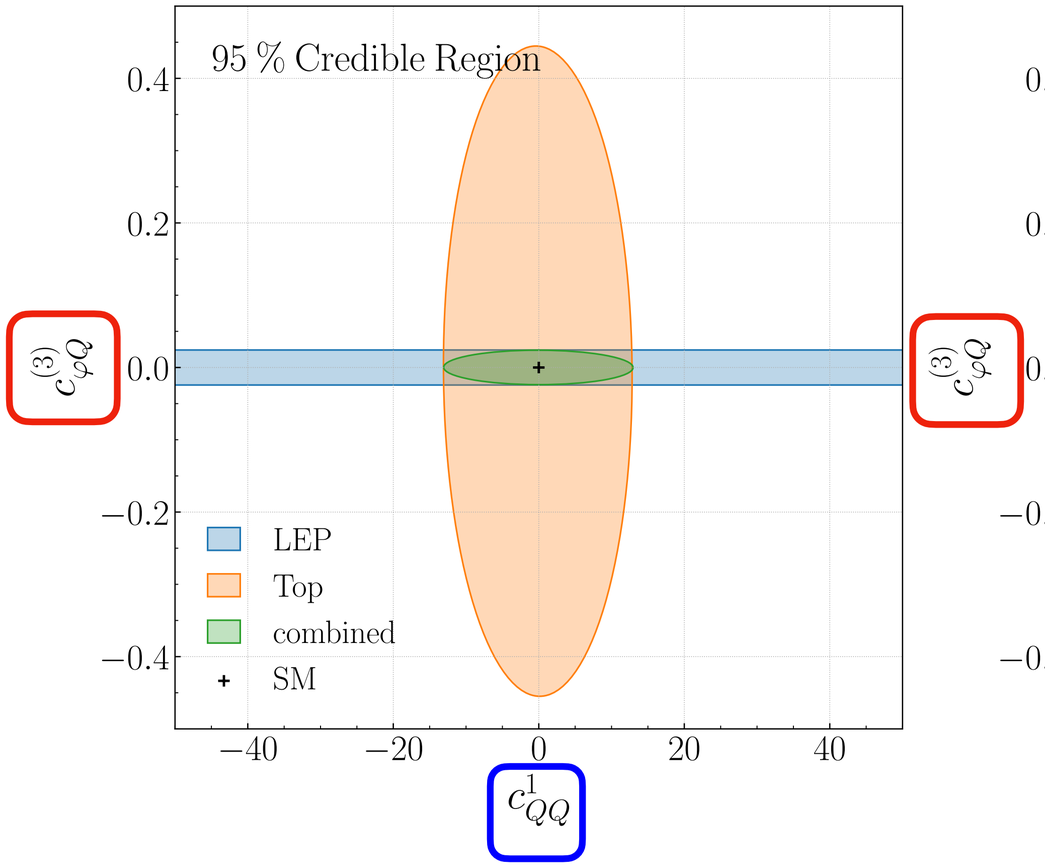
Mostly constrained from top

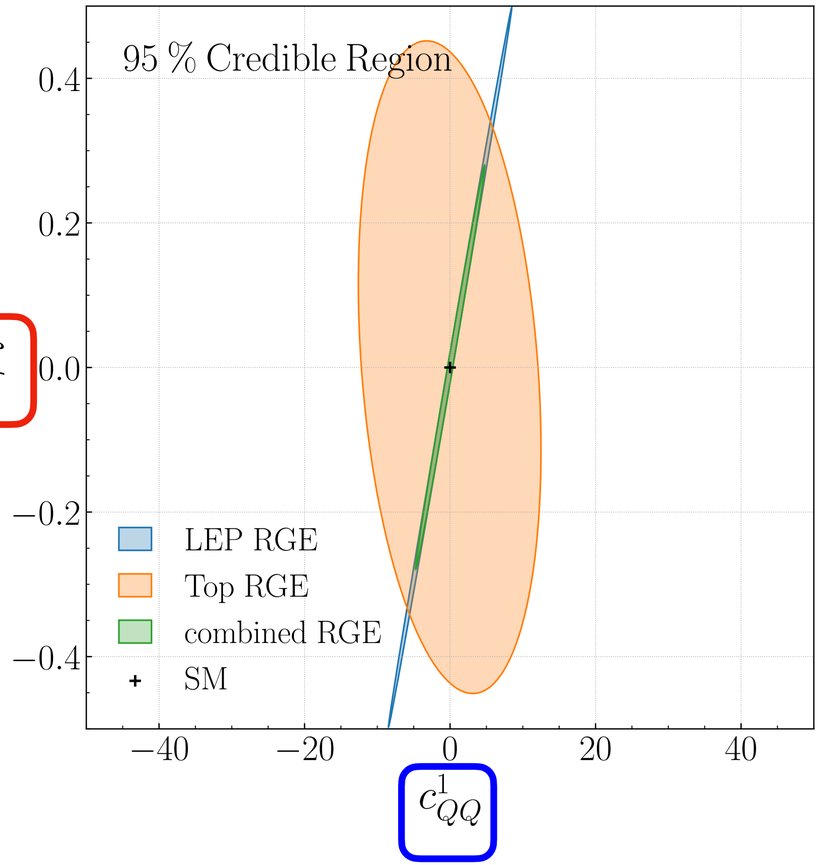


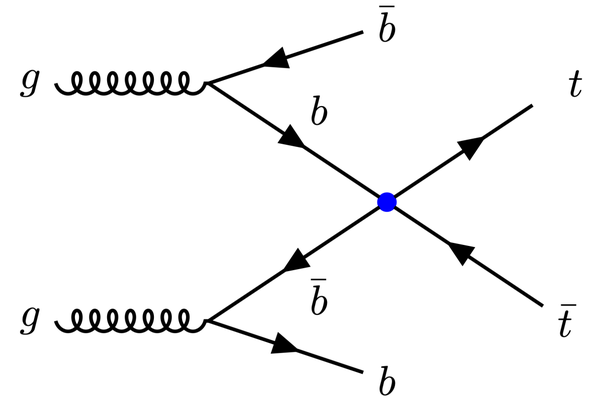
Also constrained from Higgs through $O_{\varphi B}$ and $O_{\varphi WB}$

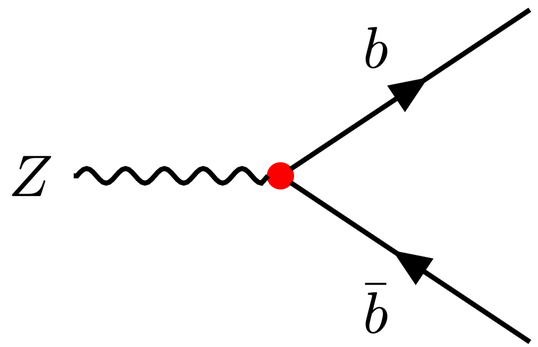
RG effects in the global fit

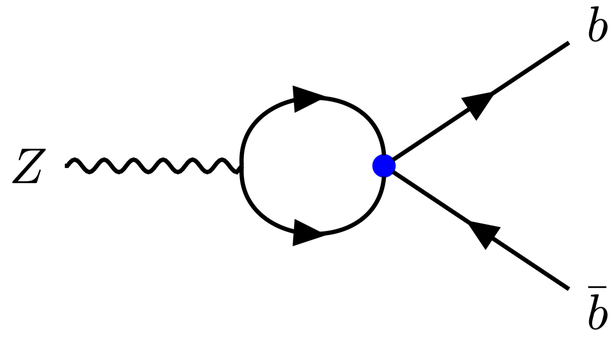
….or severely overestimate the bounds!



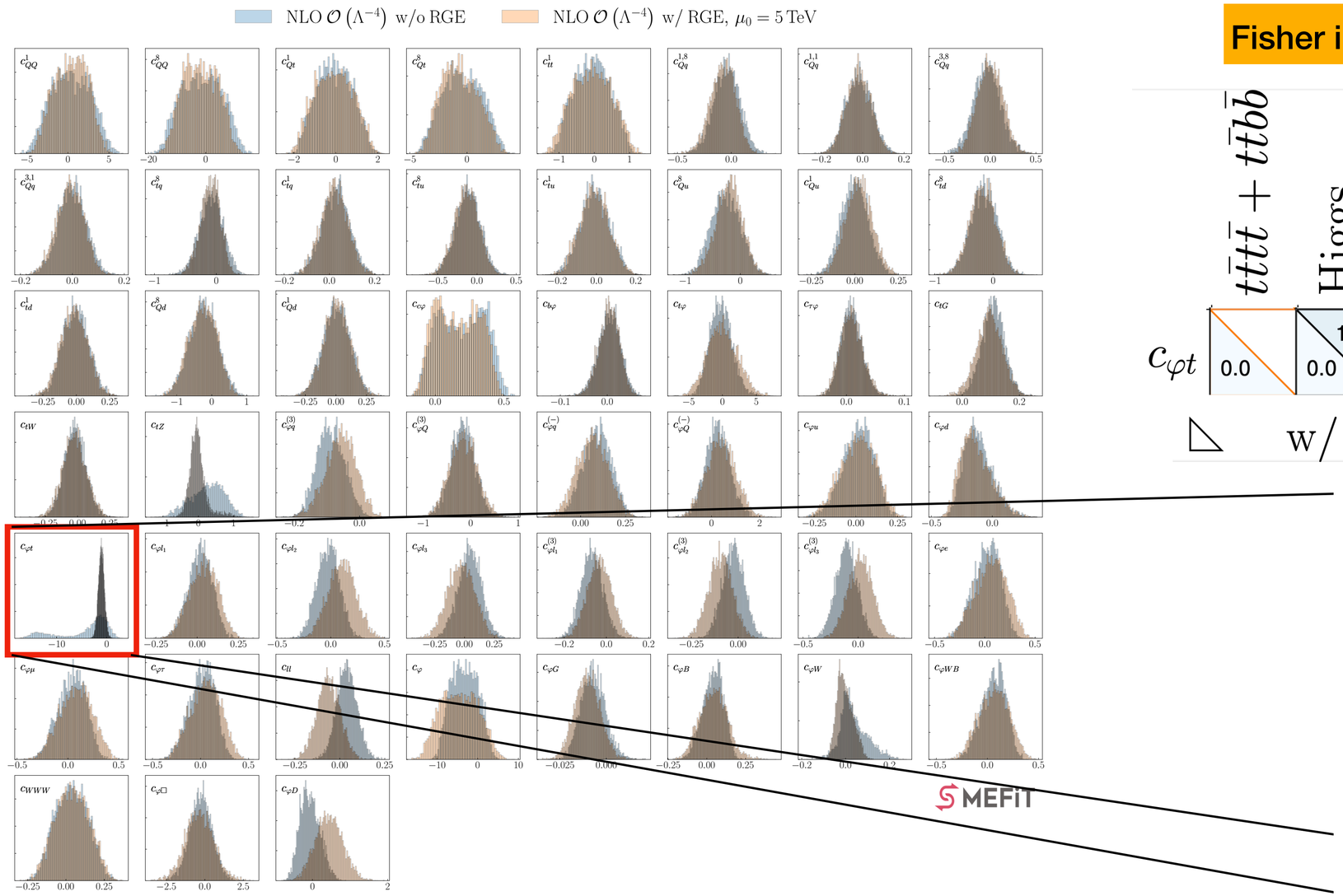




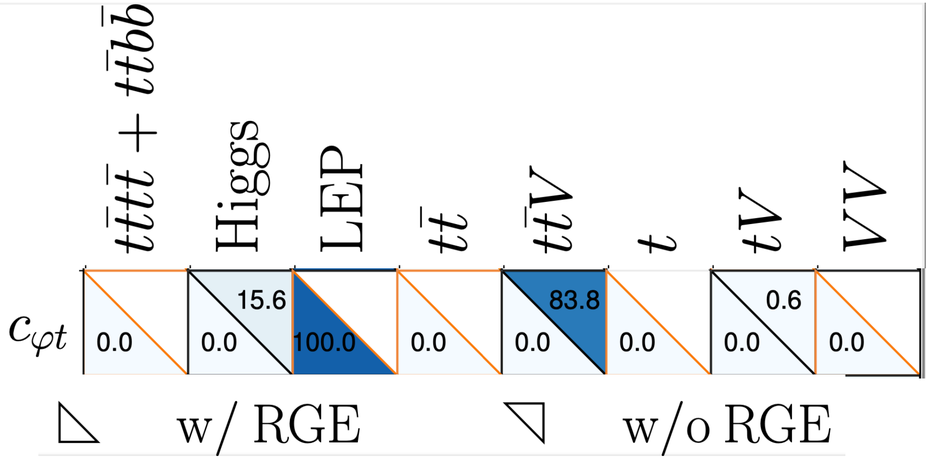


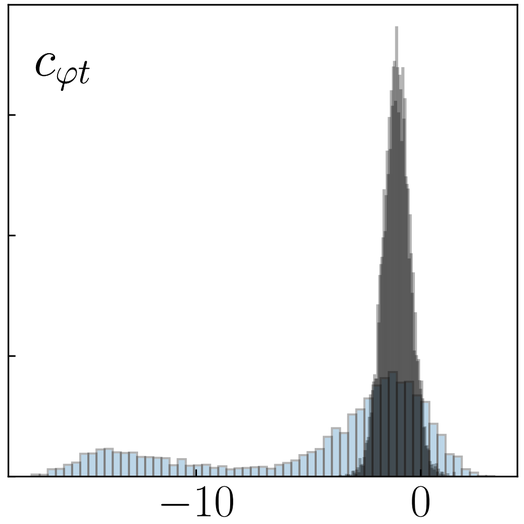


RG effects in the global fit



Fisher information moves from $t\bar{t}V$ to LEP





SMEFT at future colliders

HL-LHC projections

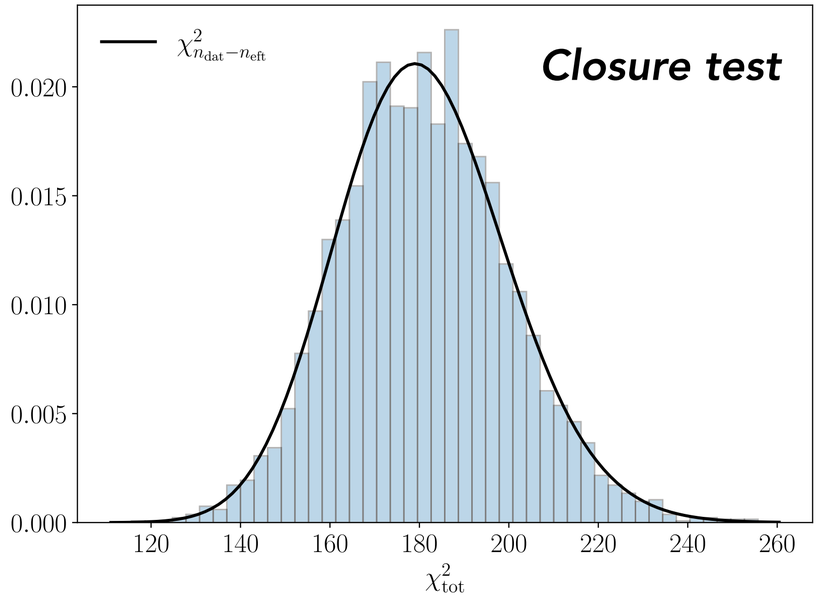
- Suppose we fluctuate the central values of the pseudo data around the SM

$$\mathcal{O}_i^{(\mathrm{exp})} = \mathcal{O}_i^{(\mathrm{th})} \left(1 + r_i \delta_i^{(\mathrm{stat})} + \sum_{k=1}^{n_{\mathrm{sys}}} r_{k,i} \delta_{k,i}^{(\mathrm{sys})} \right)$$

- while rescaling the statistical uncertainties according to the improved luminosity

$$\delta_i^{(\mathrm{stat})} = \tilde{\delta}_i^{(\mathrm{stat})} \sqrt{\frac{\mathcal{L}_{\mathrm{Run2}}}{\mathcal{L}_{\mathrm{HLLHC}}}} \qquad \delta_{k,i}^{(\mathrm{sys})} = \tilde{\delta}_{k,i}^{(\mathrm{sys})} \times f_{\mathrm{red}}^{(k)}$$

- and the systematics by an overall factor, namely 1/2 for all datasets, how do the EFT bounds change?



+ flexible framework that can project any Run II dataset

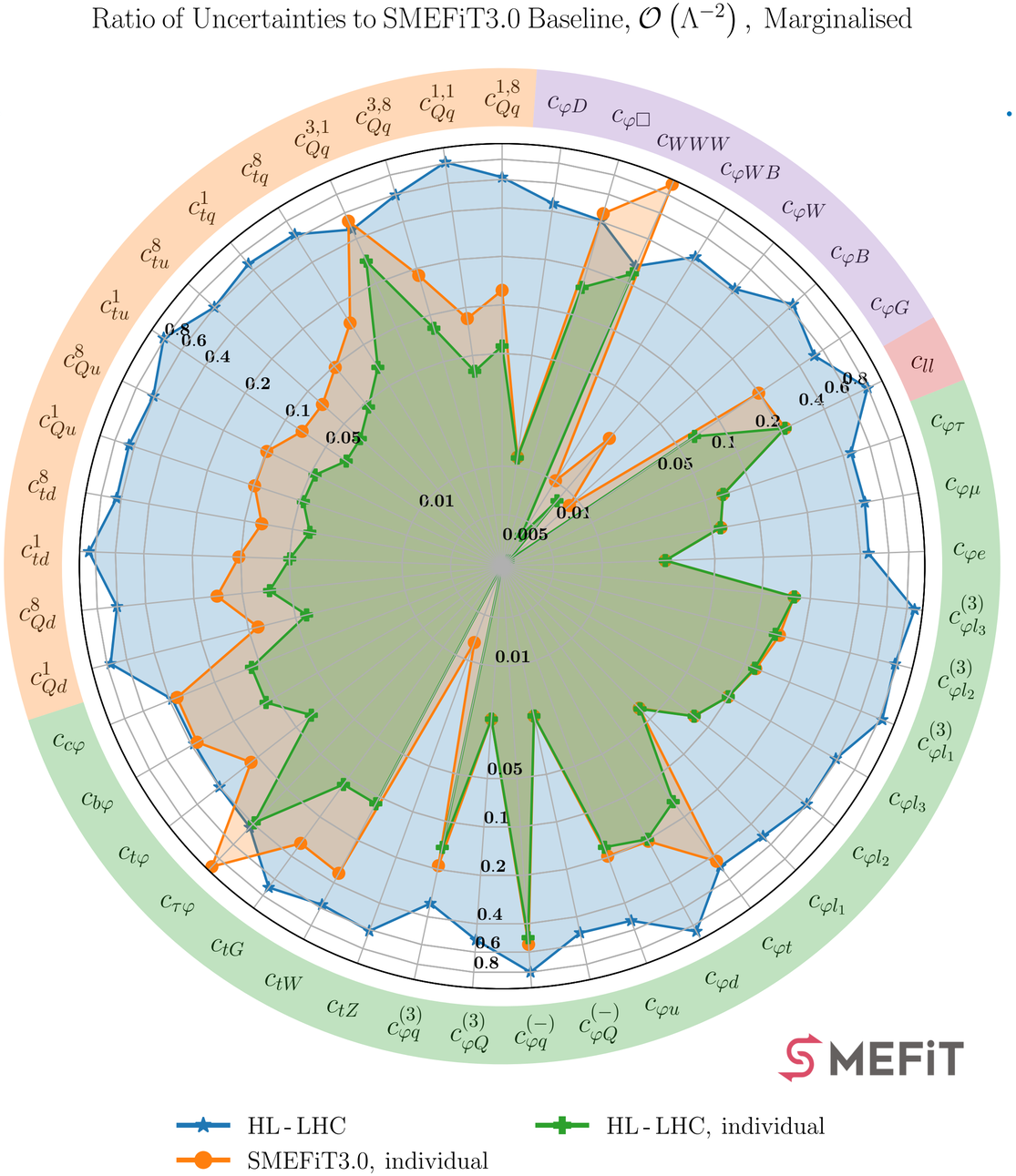
+ SMEFT predictions can be recycled

- No additional bins in the tails

The SMEFT at HL-LHC

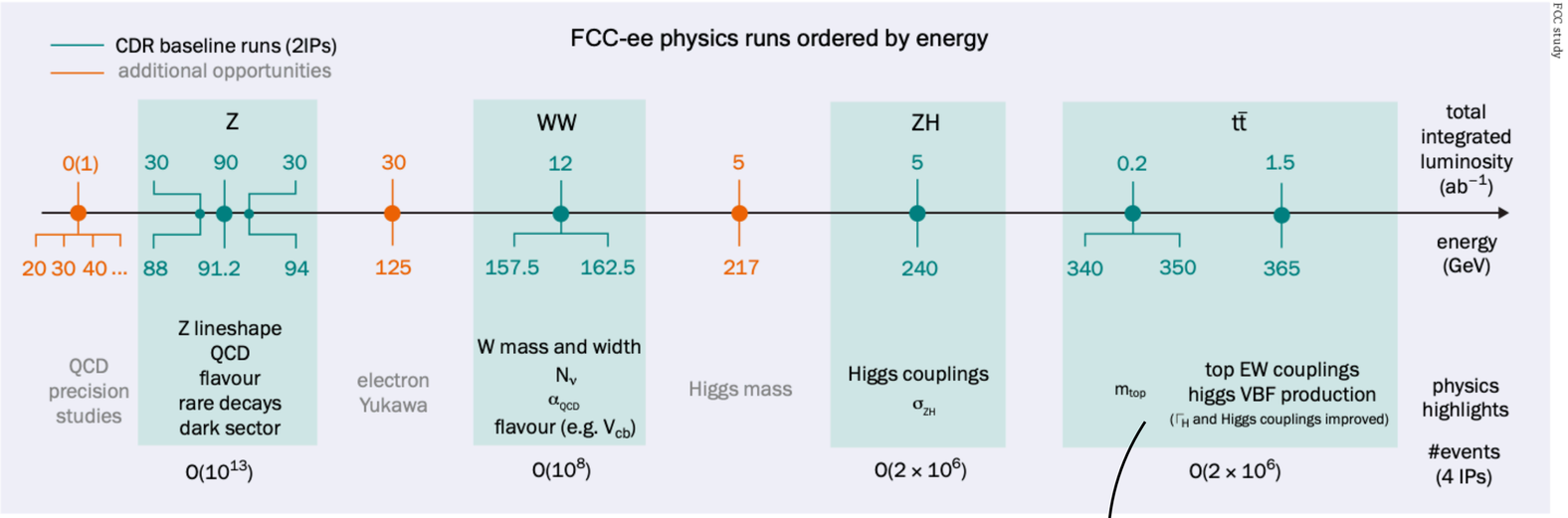
SMEFiT collaboration, arXiv: 2404.12809

- We project all RunII datasets from the SMEFiT 3.0 baseline: one for each process and final state see backup for details
- We see an improvement ranging from 20 to 70 % in the marginalised fit
- EW operators improve only in the marginalised fit



Physics programme at the FCC-ee

- At the FCC-ee, we will separately target Z-pole EWPOs, diboson, Higgs and top pair production, each with an unprecedented luminosity!
- Expected production: $6 \cdot 10^{12}$ Z bosons, $2.4 \cdot 10^{8}$ W pairs, $2 \cdot 10^{6}$ H bosons and $2 \cdot 10^{6}$ top quark pairs within 16 years

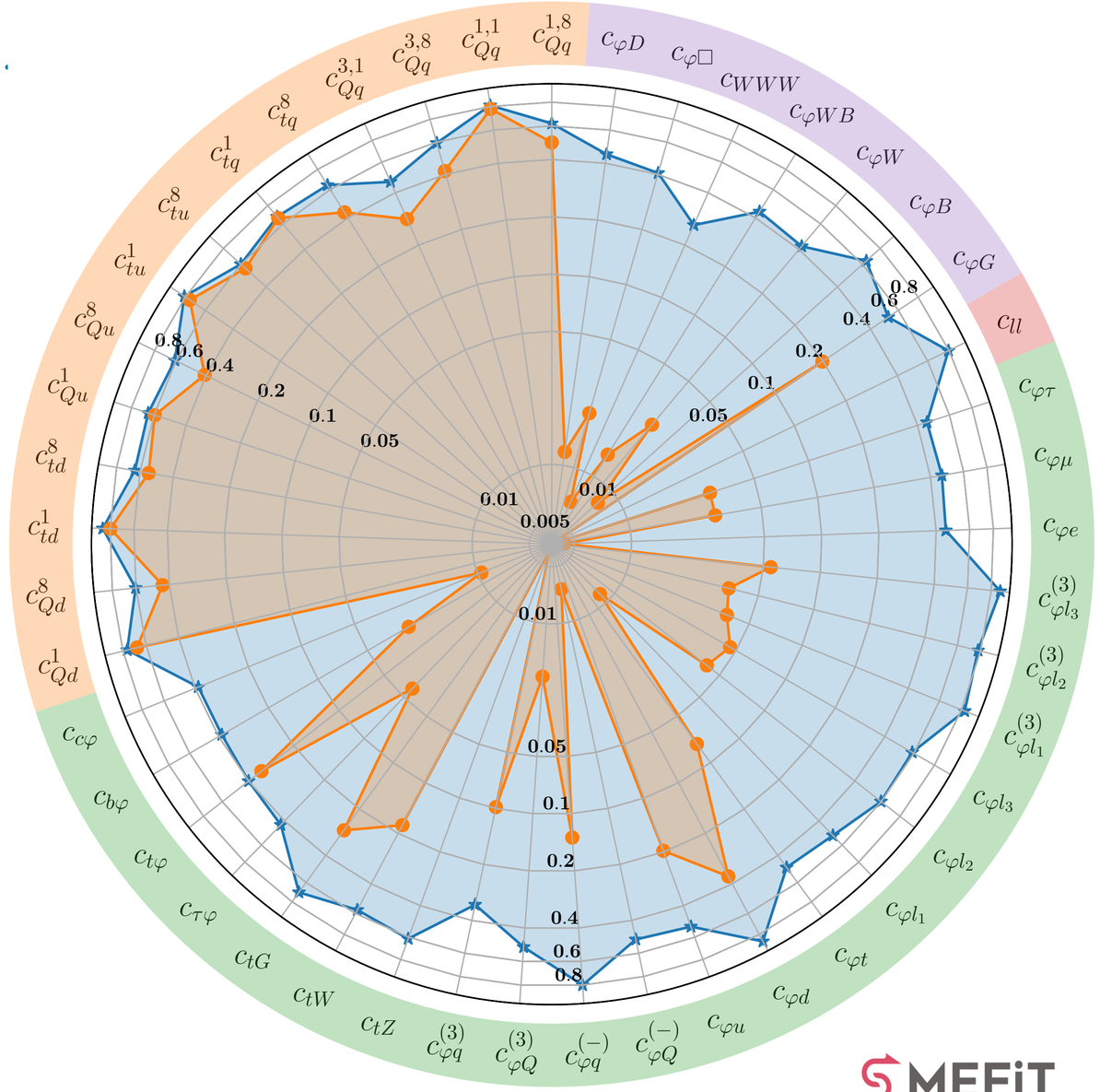


Impossible to produce with an e^+e^- collider in the LHC tunnel!

The SMEFT at FCC-ee

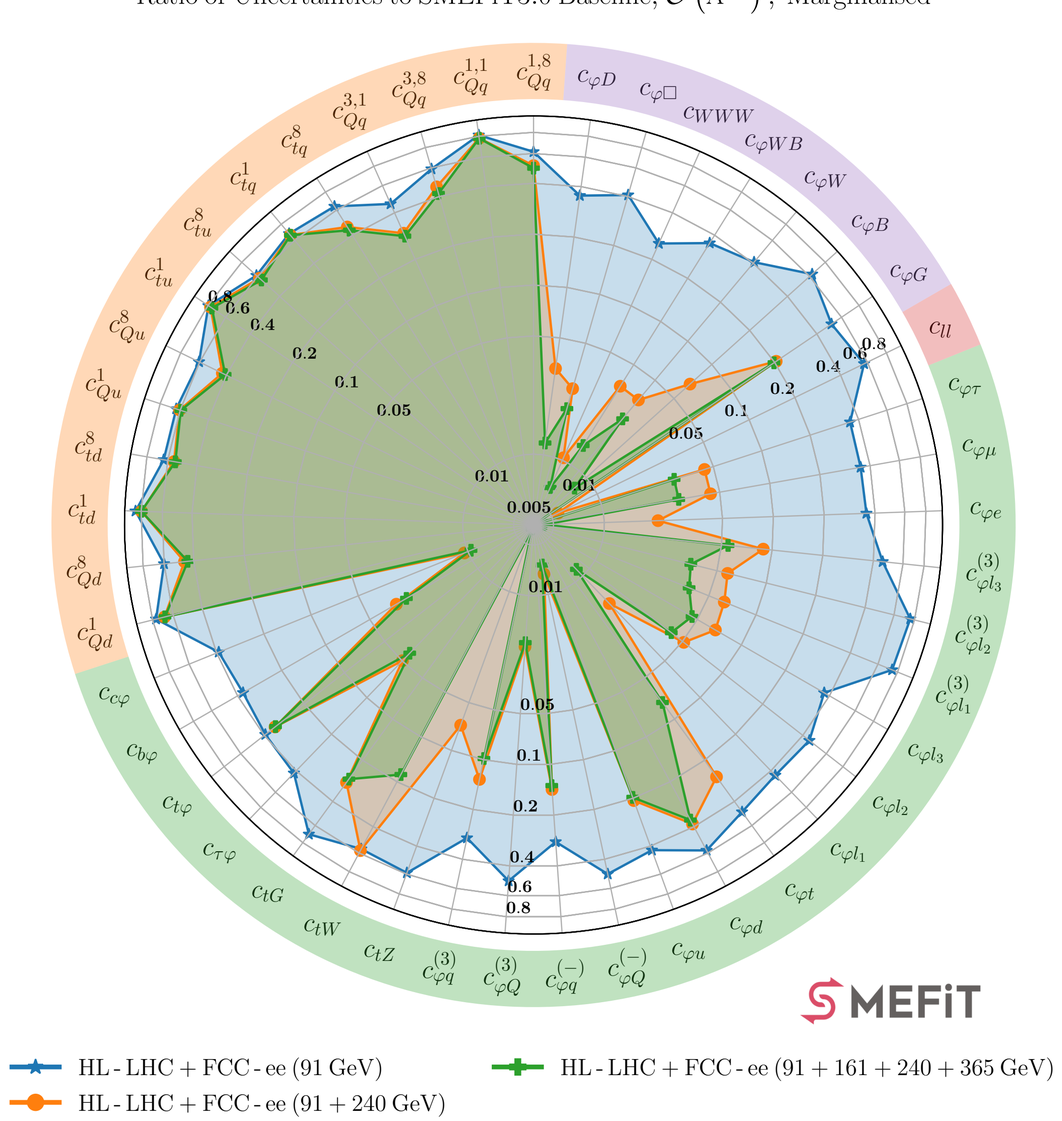
- EWPOs at the Z-pole
- Light fermion pair prediction
- Higgstrahlung and VBF
- Gauge boson pair production
- Top-quark pair production
- Optimal Observables

Energy ($\sqrt{s}$)	$\mathcal{L}_{\text{int}}$ (Run time)	
	FCC-ee	CEPC
91 GeV (Z-pole)	300 ab^{-1} (4 years)	100 ab^{-1} (2 years)
161 GeV ($2\,m_W$)	20 ab^{-1} (2 years)	6 ab^{-1} (1 year)
240 GeV	10 ab^{-1} (3 years)	20 ab^{-1} (10 years)
350 GeV	0.4 ab^{-1} (1 years)	-
365 GeV ($2\,m_t$)	3 ab^{-1} (4 years)	1 ab^{-1} (5 years)



Result: FCC-ee energy breakdown

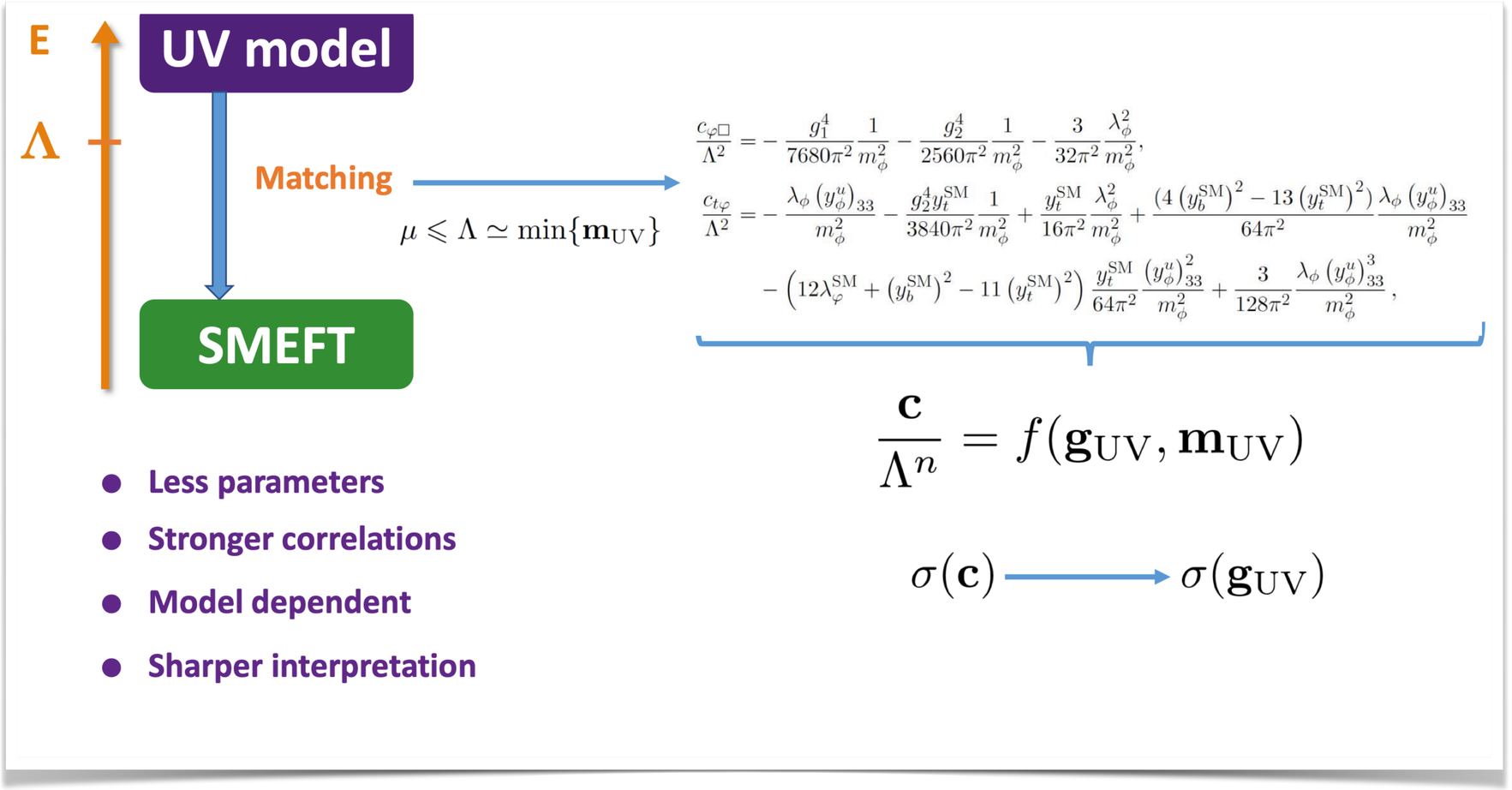
- The FCC-ee plans to operate sequentially , hence we need to study the impact at the various energies
- Largest impact for Z-pole at 91 GeV plus the Higgs factory run at 240 GeV
- We can try other combinations too in order to find the most optimal run order for the SMEFT



UV complete models

Mantani, Rojo, Rossia, Vryonidou, ***JtH***
arXiv: 2502.20453

The ultimate goal of the EFT program at the LHC is to bridge the gap to explicit UV models

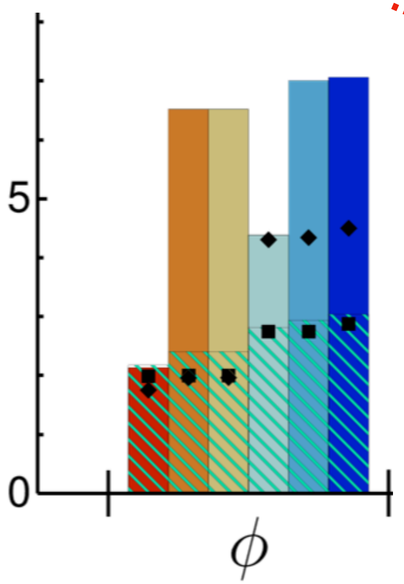


Slide from A. Rossia at SMEFT-tools 2025

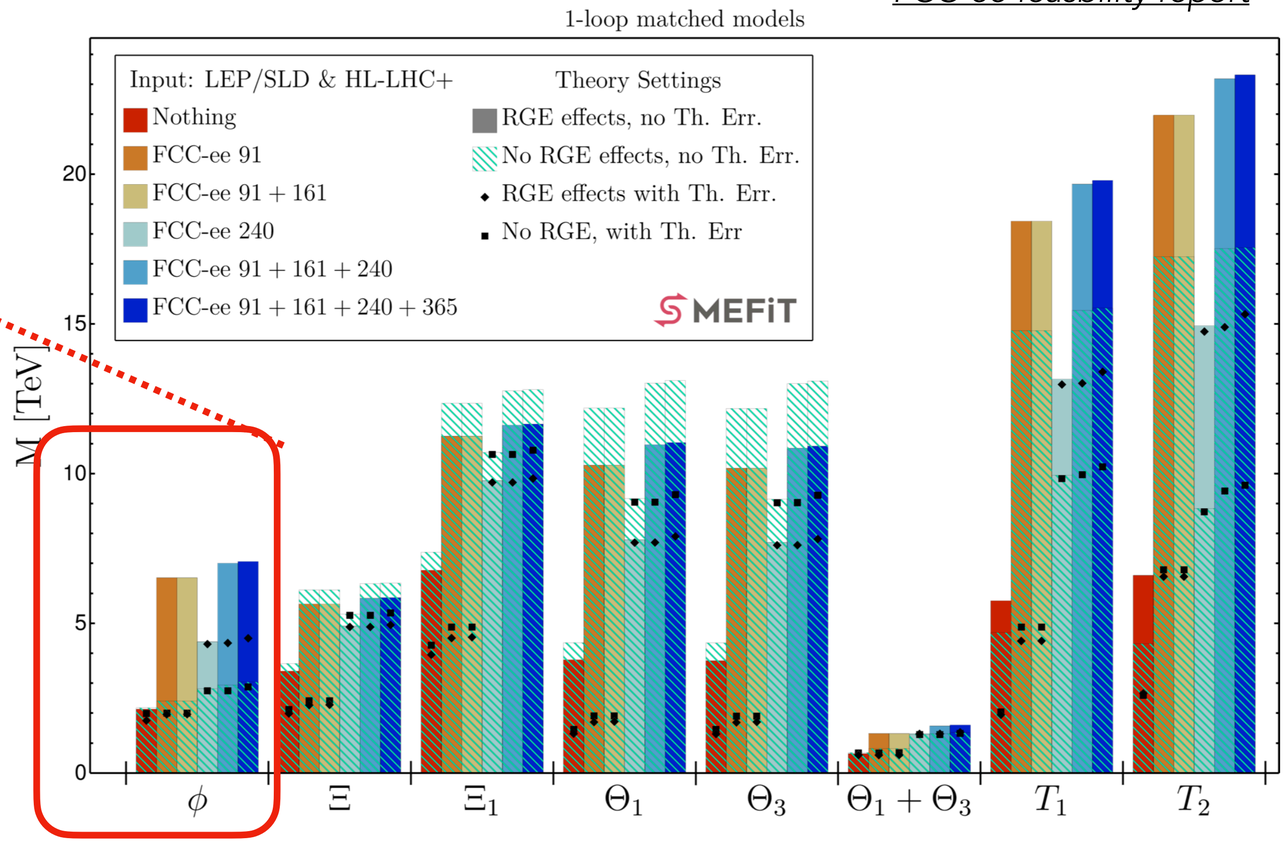
UV complete models

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RGE effects matter, and so do theory uncertainties!

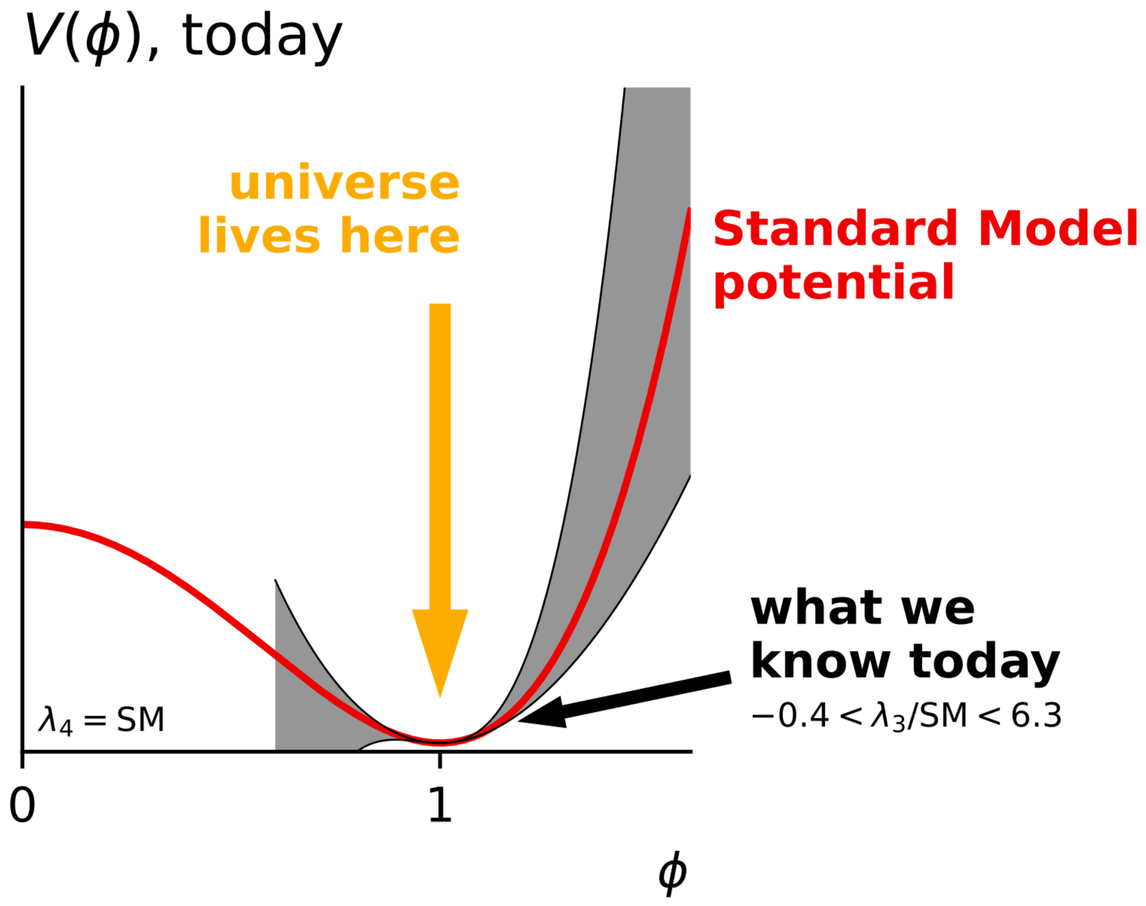


FCC-ee feasbility report



$$\mathcal{L}_{\rm UV} = \mathcal{L}_{\rm SM} + |D_\mu \phi|^2 - m_\phi^2 \phi^\dagger \phi - \Big((y_\phi^e)_{ij}\, \phi^\dagger \bar{e}_R^i \ell_L^j + (y_\phi^d)_{ij}\, \phi^\dagger \bar{d}_R^i q_L^j + (y_\phi^u)_{ij}\, \phi^\dagger i\sigma_2 \bar{q}_L^{T,i} u_R^j + \lambda_\phi\, \phi^\dagger \varphi |\varphi|^2 + \text{h.c.} \Big) - \text{scalar potential}$$

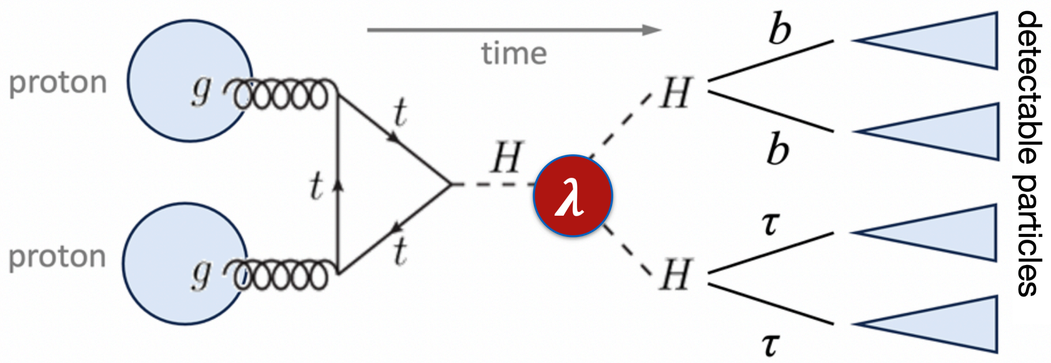
The Higgs self-coupling in the SMEFT



From Gavin Salam

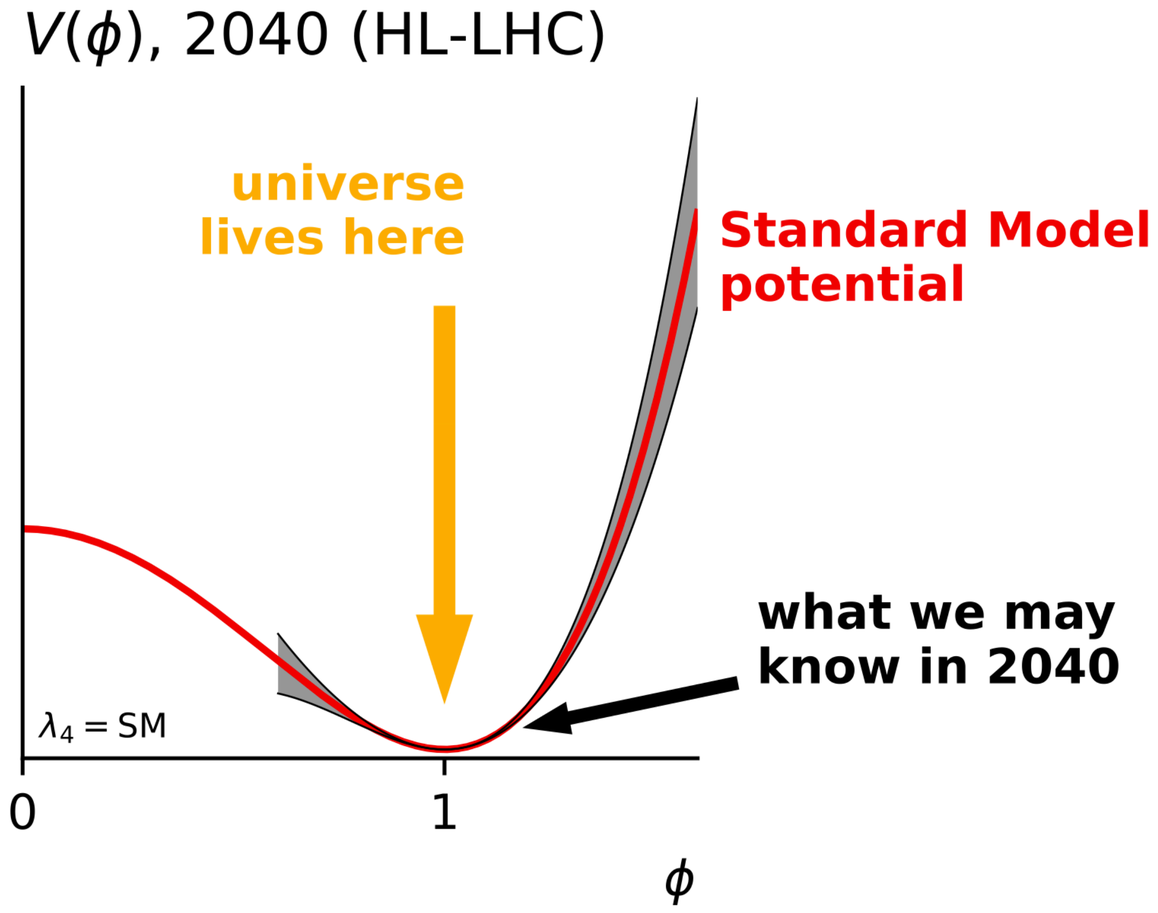
The Higgs potential sheds light on …

- the vacuum stability of our universe
- EW phase transition
- Matter anti-matter asymmetry



$$\delta\kappa_3 = -\frac{2v^4}{m_h^2}\frac{c_\varphi}{\Lambda^2} + \frac{3v^2}{\Lambda^2}\left(c_{\varphi\Box} - \frac{1}{4}c_{\varphi D}\right)$$

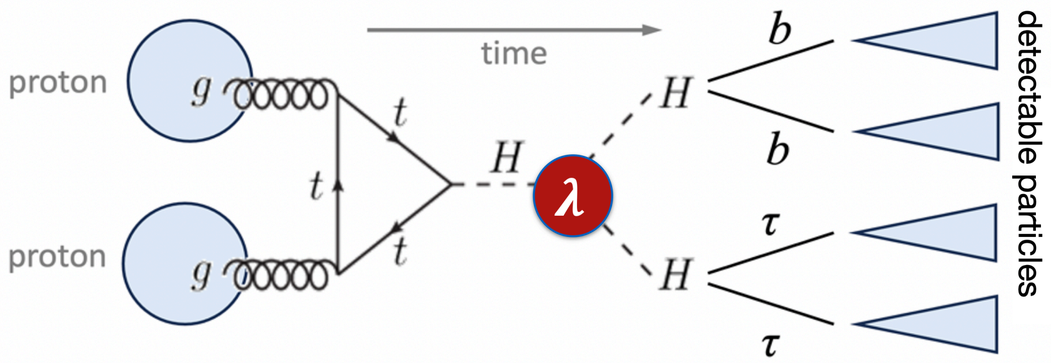
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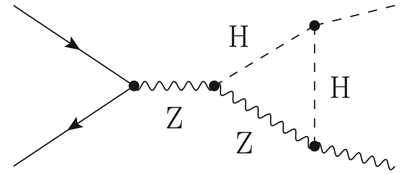


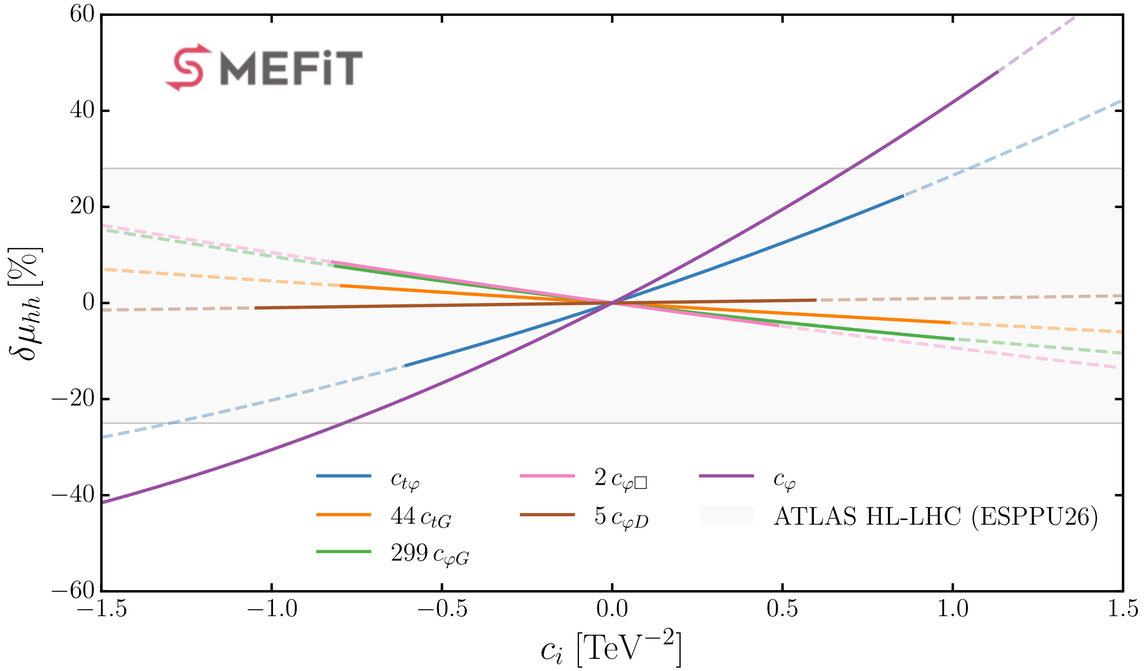
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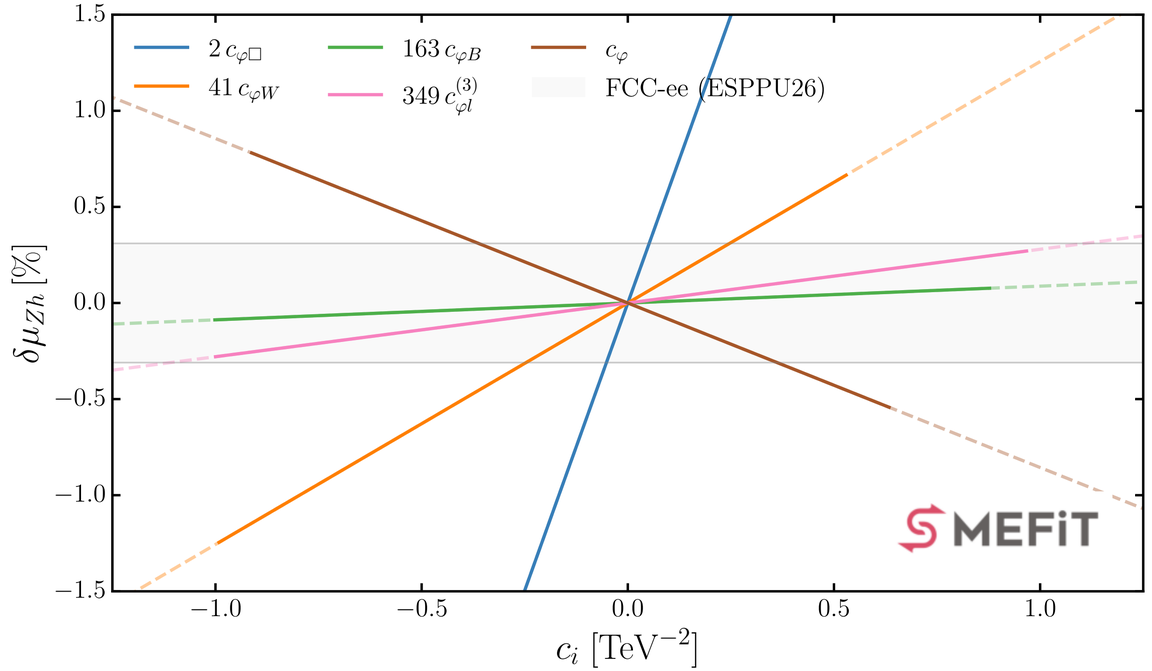
The Higgs self-coupling in the SMEFT

- Individual and marginalised bounds on c_φ are expected to be similar at HL-LHC, while the interplay with $c_{\varphi D}$ and $c_{\varphi\Box}$ becomes relevant through ZH at FCC-ee

$$\delta\kappa_3 = -\frac{2v^4}{m_h^2}\frac{c_\varphi}{\Lambda^2} + \frac{3v^2}{\Lambda^2}\left(c_{\varphi\Box} - \frac{1}{4}c_{\varphi D}\right)$$





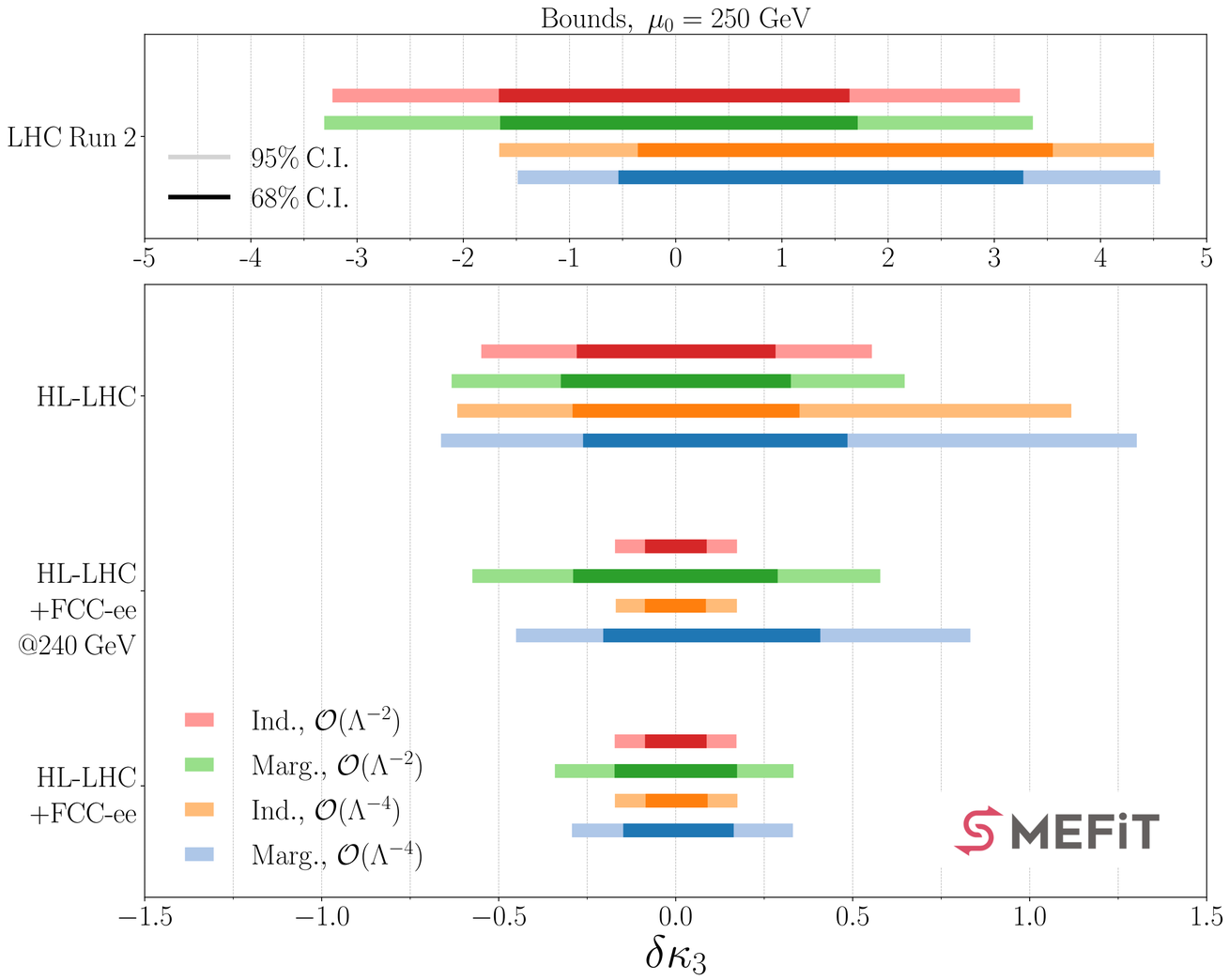


The Higgs self-coupling in the SMEFT

*Mantani, Rojo, Rossia, Vryonidou, **JtH***
arXiv: 2504.05974, submitted to PRL

- HL-LHC + FCC-ee can pin down the Higgs self coupling to 15%
- Individual and marginalised bounds on κ_3 no longer agree at FCC-ee
- Significantly improving on HL-LHC is not possible without the 365 GeV run

Input Dataset	EFT	$\delta\kappa_3$ (68% C.I.)
LHC Run 2	Linear	$[-1.68, 1.68]$
	Quad.	$[-0.54, 3.27]$
HL-LHC	Linear	$[-0.32, 0.32]$
	Quad.	$[-0.26, 0.49]$
HL-LHC & FCC-ee(240)	Linear	$[-0.29, 0.29]$
	Quad.	$[-0.20, 0.41]$
HL-LHC & FCC-ee	Linear	$[-0.17, 0.17]$
	Quad.	$[-0.15, 0.16]$



Summary and conclusion

- The SMEFT provides a convenient tool to search for new physics in an agnostic way
- The impact of RGE effects on the SMEFT parameter space is non trivial and should be included
- Complementarity between the different FCC-ee runs is key to pin down the Higgs trilinear coupling to 15%



The SMEFiT code is open source and can be installed from *lhcfitnikhef.github.io/smefit_release*

Summary and conclusion

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Contact: jaco.ter.hoeve@ed.ac.uk **Thanks for your attention!**